Angular persistence and the universal pre-asymptotic Hurst exponent of thermal soaring flights

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Abstract. A recent empirical analysis [J. Vilpellet, A. Darmon and M. Benzaquen, arXiv:2601.01293 (2026)] reports a sub-ballistic Hurst exponent $H \approx 0.88$ for horizontal transport in paraglider, hang glider and sailplane flights that is essentially independent of aircraft type, despite sizeable differences in characteristic horizontal speed and in the Pareto-tail exponents of the transition- and search-phase duration distributions. We introduce a minimal cycle-based continuous-time random walk (CTRW) model in which each elementary step is a complete transition–search–climb cycle. Phase durations are i.i.d. with Pareto tails for transition and search (tail exponents $\mu_{\mathrm{T}}, \mu_{\mathrm{S}} > 1$ taken from the reference analysis) and an exponential law for climb. Each phase carries its own intra-phase horizontal dynamics, independently specified: a ballistic translation during transition; an intermittent two-state motion during search, in which exponential-duration ballistic repositioning legs alternate with Mittag–Leffler-distributed localised scouting rests; and a 2-D harmonic oscillator with per-cycle turn-period dispersion plus a weak orographic drift during climb. Successive transition headings evolve by a Gaussian random walk on the circle with per-cycle variance σ_θ^2. A closed-form expression for the discrete-step mean-square displacement (MSD) is derived, exhibiting a pre-asymptotic crossover window in which a single power-law fit yields effective Hurst exponents in $(1/2, 1)$. The $(\mu_{\mathrm{T}}, \sigma_\theta)$ phase diagram of the model places the iso-line $H_{\mathrm{eff}} = 0.88$ at $\sigma_\theta \simeq 0.35\,\mathrm{rad}$ essentially independently of μ_{T}, supplying an analytical origin for the observed universality. Monte Carlo simulations at this universal σ_θ reproduce the empirical MSD with a tight aircraft-to-aircraft spread $\Delta H \lesssim 0.01$: paragliders $H = 0.934$, hang gliders $H = 0.930$, sailplanes $H = 0.937$. Rescaling by v_{xy}^2 collapses the three curves onto a single master function of the lag, reproducing the universal collapse reported experimentally (inset of Fig. 1 of the reference analysis). The paraglider intra-phase parameters are calibrated on Fig. 3 of the reference analysis; hang-glider and sailplane Pareto scales $\tau_0^{\mathrm{T}}, \tau_0^{\mathrm{S}}$ (which the reference analysis does not publish) are tuned so that $\langle \tau^{\mathrm{T}} \rangle$ and $\langle \tau^{\mathrm{S}} \rangle$ match the paraglider values, thereby collapsing $\delta^2(\Delta)/v_{xy}^2$; σ_θ is imposed analytically from the phase diagram rather than fitted.

1 Introduction

Anomalous transport in heterogeneous, patchy, and intermittent environments is a recurrent theme in physics, biology and ecology [1–4]. Cross-country soaring flight provides a natural instance: paragliders, hang gliders and sailplanes exploit atmospheric thermal updrafts as localised energy patches, alternating ballistic glides between thermals with localised circling motion inside them. Large GPS trajectory archives have recently enabled population-scale statistical analyses of the resulting transport [5].

In a comprehensive empirical study, Vilpellet, Darmon and Benzaquen [5] analysed 49 939 paraglider, 780 hang-glider and 9 203 sailplane trajectories and reported a striking observation: beyond a short ballistic regime ($\Delta \lesssim 10\,\mathrm{s}$), the horizontal MSD grows as $\delta^2(\Delta) \propto \Delta^{2H}$ with $H \approx 0.88$ for *all three* aircraft types. This is non-trivial in view of the differences in horizontal velocity v_{xy} (factor ~ 3.6 between paragliders and sailplanes) and in the empirical tail exponents of the phase-duration distributions (factor ~ 1.8 in μ_T, see Table 1).

Using a hidden Markov model [6], the authors of [5] segmented each flight into repeating transition–search–climb cycles and showed that within each phase the motion has well-defined but very different

scaling: transition phases are individually ballistic, search phases are strongly subdiffusive, and climb phases produce very little net horizontal displacement — approximately two orders of magnitude below v_{xy} according to [5] — because thermal updrafts are typically tilted by ambient wind and breeze, causing a slow drift rather than true vertical motion.

In their Discussion, Vilpellet *et al.* [5] discuss a Lévy-walk-type mechanism [2] as a candidate explanation, noting that in this regime one would expect $\delta^2 \sim \Delta^{4-\mu}$ (density-convention formula) and that the observed $H = 0.88$ would correspond to $\mu \approx 2.25$. They note however that the measured μ_T values differ across aircraft while the Hurst exponent does not, incompatible with a naive aircraft-by-aircraft Lévy-walk interpretation. They conclude that "two-dimensional directional decorrelation" and "pre-asymptotic scaling" [1, 7] must also play a role but leave the quantitative modelling as an open problem. In addition, the Fig. 4 caption of [5] declares the survival convention $P(\tau > t) \sim t^{-\mu}$, whereas the $\Delta^{4-\mu}$ formula and the statement "$\mu < 3$, infinite variance" in their text tacitly assume the density convention $f(\tau) \sim \tau^{-\mu}$, an internal ambiguity we address explicitly (Appendix A).

In this Letter we show that a minimal cycle-based CTRW model with an explicit angular-persistence mechanism reproduces the observed universality quantitatively. We derive a closed-form expression for the MSD after N cycles (App. C); characterise a pre-asymptotic crossover window — the range of time lags in which the system has not yet reached its diffusive asymptotics — over which a single power-law fit yields an effective exponent $H_{\text{eff}} \in (1/2, 1)$; and map out the (μ_T, σ_θ) phase diagram of H_{eff}. The phase diagram exhibits nearly horizontal isocurves of H_{eff} (Fig. 7) and locates the empirical $H = 0.88$ contour at $\sigma_\theta \simeq 0.35\,\text{rad}$, essentially independent of μ_T; we identify this weak μ_T-dependence as the analytical origin of the universality. A direct Monte Carlo simulation of the full model with this angular diffusivity, using paragliders intra-phase parameters tuned on Fig. 3 of [5] and the parameters of Table 1 for the other two aircraft classes, yields $H = 0.934, 0.930, 0.937$ respectively (fitted in $10\,\text{s} < \Delta < 5 \times 10^3\,\text{s}$). Rescaling by v_{xy}^2 collapses the three MSD curves onto a single master function of the lag (Fig. 3), reproducing the universal transport reported experimentally in the inset of Fig. 1 of [5].

	paragliders	hang gliders	sailplanes
v_{xy} (m/s)	10.06	15.58	35.85
μ_{T} (transition)	3.93	4.79	2.62
μ_{S} (search)	3.88	2.25	2.90
μ_{C} (climb)	8.28	5.33	4.54
η_{T} (trans. fraction)	0.61	0.56	0.58
η_{S} (search fr.)	0.06	0.05	0.015
η_{C} (climb fr.)	0.32	0.39	0.40
H_{obs}		0.88 (universal)	

Table 1: Empirical parameters from [5]. Tail exponents μ of phase durations are Hill-estimator values reported in the legend of Fig. 4c,f,i of the cited paper; under our convention (Appendix A) $\langle\tau\rangle < \infty$ requires $\mu > 1$ and $\langle\tau^2\rangle < \infty$ requires $\mu > 2$, so all aircraft have finite variance of phase durations. Time fractions η are ensemble means read from the pie charts in Fig. 7 of [5]. The Pareto scales $\tau_0^{\text{T}}, \tau_0^{\text{S}}$ of the phase-duration distributions are model-introduced and not published in [5]: for paragliders we take physically plausible values (see configs), for hang gliders and sailplanes we tune them so that $\langle\tau^{\text{T}}\rangle$ and $\langle\tau^{\text{S}}\rangle$ match the paraglider values, aligning the mean cycle duration $\langle T\rangle$ across aircraft and collapsing $\delta^2(\Delta)/v_{xy}^2$ onto the paraglider curve (Fig. 3). This alignment induces minor deviations ($\lesssim 0.1$) from the η_X values quoted here.

2 The cycle-based CTRW model

We regard each soaring cycle $n \in \mathbb{N}$ as a single discrete step of a two-dimensional CTRW [1, 8]. The empirical segmentation of [5] resolves each flight as a repeating ordered sequence *climb* $\to$ *transition* $\to$ *search*; after the first search only, every subsequent search is always preceded by a completed

climb. Only the *transition* phase contributes a net horizontal step in our cycle-based decomposition, the other two phases being modelled as locally reshuffled motions that carry no net persistent direction (assumption (A4) below). The horizontal position at the end of cycle n is therefore

$$\mathbf{x}_n = \mathbf{x}_{n-1} + \ell_n\, \hat{\mathbf{e}}(\theta_n), \qquad \hat{\mathbf{e}}(\theta) \equiv (\cos\theta, \sin\theta), \tag{1}$$

with step length $\ell_n = v_{xy}\, \tau_n^{\mathrm{T}}$ set by the transition duration. The cycle duration is $T_n = \tau_n^{\mathrm{T}} + \tau_n^{\mathrm{S}} + \tau_n^{\mathrm{C}}$, and the cycle-count process is $N(t) = \max\{n : \sum_{k\le n} T_k \le t\}$.

The model is fully specified by the following assumptions, each an explicit modelling choice on top of the CTRW framework itself:

(A1) *Cycle statistics.* The triples $\{(\tau_n^{\mathrm{T}}, \tau_n^{\mathrm{S}}, \tau_n^{\mathrm{C}})\}_n$ are i.i.d. with independent marginals. The empirical survival functions of Fig. 4c,f,i of [5] display heavy power-law tails for transition and search and a light exponential tail for climb; accordingly we parametrise

$$P(\tau^X > \tau) = \left(\tau_{\min}^X/\tau\right)^{\mu_X}, \quad \tau \ge \tau_{\min}^X,\ X \in \{\mathrm{T}, \mathrm{S}\}, \tag{2}$$

(a Pareto tail in the survival convention of Fig. 4 of [5]), and $\tau^{\mathrm{C}} \sim \mathrm{Exp}(\bar{\tau}_{\mathrm{C}})$, with tail exponents μ_X taken directly from the Hill fits of Fig. 4c,f,i of [5] (Table 1). The scale parameters $\tau_{\min}^X$ are modelling choices (not read from [5]) set so that the resulting mean durations sit in a realistic range; see the code configuration files. All empirical μ_X are > 1 (Table 1), so $\langle \tau^X \rangle$ is finite; the more general Mittag–Leffler CTRW framework [1, 8], valid for $\mu_X \in (0, 1)$, thus reduces here to its Pareto tail, and we work with Eq. (2) directly. The Mittag–Leffler law itself plays a different role in the model, deferred to the search-phase dynamics (A2.S) below.

(A2) *Intra-phase horizontal motion.* During the transition phase of cycle n the aircraft moves deterministically at velocity $v_{xy}\, \hat{\mathbf{e}}(\theta_n)$, producing a ballistic MSD $\delta_T^2(\Delta) = v_{xy}^2 \Delta^2$ at all lags below τ_n^T. The search and climb phases are modelled differently because their empirical MSDs in Fig. 3 of [5] display qualitatively different features:

(A2.S) Search: intermittent ballistic CTRW with Mittag-Leffler rests. Inside each search phase of duration τ_n^S, the pilot's horizontal motion in the xy-plane alternates between two physically distinct regimes:

1. *Ballistic legs* of duration $\sigma_k \sim \mathrm{Exp}(\sigma_0)$, during which the walker translates at effective speed $v_c^S > v_{xy}$ in the xy-plane along a persistent direction ψ_k (intra-phase angular Gaussian random walk on the circle, $\psi_k = \psi_{k-1} + \zeta_k$, $\zeta_k \sim \mathcal{N}(0, \sigma_\psi^2)$; not to be confused with the cycle-to-cycle angular increment $\eta_n \sim \mathcal{N}(0, \sigma_\theta^2)$ of (A3) below).
2. *Waiting epochs* of duration ξ_k drawn from a Mittag-Leffler distribution,

$$P(\xi_k > \xi) = E_{\alpha_S}\left[-(\xi/\tau_0^S)^{\alpha_S}\right], \tag{3}$$

with $\alpha_S \in (0, 1)$, during which the walker's horizontal position does not advance on the scale of the GPS sampling.

This two-component dynamics is a *CTRW with finite-velocity (ballistic) steps of duration σ_k and Mittag–Leffler waiting times ξ_k* [1, 8]. Because σ_k is exponentially distributed the individual steps are not heavy-tailed, and this process is therefore *not* a Lévy walk [2] (whose legs would be power-law in duration): the sub-diffusive long-time behaviour arises entirely from the heavy-tailed waits ξ_k. The CTRW admits a rigorous continuum-limit representation as a subordinated process,

$$\mathbf{X}^S(t) \;=\; \tilde{\mathbf{X}}^S\big(E_{\alpha_S}(t)\big) \quad \text{(in distribution, at } t \gg \tau_0^S), \tag{4}$$

where $\tilde{\mathbf{X}}^S(\cdot)$ is the leg-count-indexed displacement process and $E_{\alpha_S}(t)$ is the inverse of a one-sided α_S-stable subordinator, the continuum limit of the Mittag–Leffler renewal counting process. A self-contained derivation — Laplace-transform analysis of the renewal density and the Meerschaert–Scheffler [9] limit theorem — is given in Appendix H.

Physical interpretation of the waiting times (modelling hypothesis). The following is a *modelling interpretation*, not a feature extracted from the data of [5]: the raw trajectories were segmented into three phases via a hidden Markov model [6] and no further sub-segmentation of the search phase is reported there. We propose the following reading of the two-component CTRW of (A2.S). The pilot is *never strictly at rest* — a paraglider, hang glider or sailplane cannot hover. We therefore hypothesise that the waiting epochs ξ_k of (A2.S) correspond to the horizontal projection of a *localised circling motion* that the pilot performs while probing a candidate updraft. Under this hypothesis, during each "tight-scouting" epoch the pilot circles in place at an effective radius $r^{\text{scout}} \ll r_0^{\text{climb}}$ (assumed of order a few metres) with a short turn period (assumed of order a few seconds), much faster than the thermalling turn of the climb phase. In the xy-projection at GPS sampling rate ($\Delta t \sim 1\,\text{s}$), such tight circling would produce a position signal fluctuating at an amplitude below the effective spatial resolution of the sampled trajectory, and would therefore be observationally indistinguishable from a pause — consistent with the HMM output but not directly confirmed by it. The broad Mittag-Leffler distribution of the ξ_k is then a further hypothesis, reflecting a broad range of scouting times invested on different candidate spots (from quick discard, $\xi \sim 1\,\text{s}$, to prolonged evaluation, $\xi \sim$ minutes); the empirical intra-search MSD of Fig. 3e of [5] is consistent with this heavy-tailed waiting-time ansatz but does not by itself pin down the full distribution.

The ballistic legs, in turn, are interpreted — again as a modelling hypothesis — as short tactical manoeuvres, i.e. repositioning glides between neighbouring scouting spots. Their effective speed $v_c^S > v_{xy}$ is taken so that during a pure repositioning glide the pilot, not circling, moves faster in the xy-plane than the cycle-averaged horizontal speed v_{xy} (which incorporates the tight-circling periods). The exponential law for σ_k is chosen for minimality (memoryless manoeuvres); it is not a direct empirical observation of individual leg durations, which are not resolved separately in the HMM output.

With Mittag-Leffler waits and exponential legs, the MSD of (A2.S) has three regimes separated by the two timescales σ_0 and τ_0^S:

$$\delta_S^2(\Delta) \sim \begin{cases} (v_c^S)^2\,\Delta^2 & \Delta \ll \sigma_0 \\ \text{transient} & \sigma_0 \ll \Delta \ll \tau_0^S \\ \frac{(v_c^S \sigma_0)^2}{\Gamma(1+\alpha_S)} \left(\frac{\Delta}{\tau_0^S}\right)^{\alpha_S} & \Delta \gg \tau_0^S, \end{cases} \tag{5}$$

where the first regime is the ballistic short-time inertial motion inside a single leg and the third is the asymptotic sub-diffusive scaling of the Mittag-Leffler CTRW, recovering the empirical exponent $\alpha_S = 0.6$ (paragliders, hang gliders) or $\alpha_S = 0.4$ (sailplanes) reported in Fig. 3 of [5]. The transient regime has an effective exponent between 1 and α_S, depending on σ_0/τ_0^S. A derivation of the three regimes by direct ensemble averaging of the leg-count renewal process is given in App. G.

(A2.C) Climb: 2D harmonic oscillator plus drift with per-cycle turn-period dispersion. Inside each climb phase of duration τ_n^C, the pilot circles a thermal core at a fixed aircraft-specific radius r_0 with an angular frequency $\omega_{0,n} = 2\pi/T_{\text{turn},n}$ that fluctuates cycle-to-cycle,

$$T_{\text{turn},n} \sim \mathcal{N}(\bar{T}, \sigma_T^2), \quad \text{clipped at } T_{\text{turn},n} \geq 0.2\,\bar{T}, \tag{6}$$

with $(\bar{T}, \sigma_T) = (60, 35)\,\text{s}$ for paragliders and hang gliders and $(\bar{T}, \sigma_T) = (30, 5)\,\text{s}$ for sailplanes

in the configurations used in Figs. 4–6. The circling is superposed on a linear orographic drift of magnitude $v_{\text{drift}} \ll v_{xy}$:

$$\mathbf{x}_n^C(c) = r_0 \left[\hat{\mathbf{R}}(\omega_{0,n}c + \phi_{0,n}) - \hat{\mathbf{R}}(\phi_{0,n})\right] + v_{\text{drift}}\, c\, \hat{\mathbf{e}}(\phi_n^{\text{drift}}), \tag{7}$$

where $\hat{\mathbf{R}}(\phi) = (\cos\phi, \sin\phi)$, $\phi_{0,n}$ is an initial circling phase uniform on $[0, 2\pi)$, and ϕ_n^{drift} is a drift direction drawn i.i.d. uniformly per cycle. The conditional MSD *at fixed* $T_{\text{turn},n}$ (derived in Sec. 3 below),

$$\delta_C^2(\Delta \mid \omega_{0,n}) = 2r_0^2[1 - \cos(\omega_{0,n}\Delta)] + v_{\text{drift}}^2\, \Delta^2, \tag{8}$$

averaged over the Gaussian dispersion Eq. (6), yields

$$\delta_C^2(\Delta) = 2r_0^2\left[1 - \langle\cos(\omega_0\Delta)\rangle_T\right] + v_{\text{drift}}^2\, \Delta^2, \tag{9}$$

so that the per-cycle ω_0 dispersion *damps* the coherent oscillations of the deterministic-ω_0 version beyond the first cycle. The MSD exhibits the three regimes of Fig. 3(a)-(c) of [5]: ballistic at $\Delta \ll 1/\bar{\omega}_0$ (tangent speed $v_t = r_0\bar{\omega}_0$), a single residual anti-correlation dip at $\Delta \sim \pi/\bar{\omega}_0$ (visible for the largest r_0), and a quasi-ballistic drift-dominated tail at $\Delta \gg 1/\bar{\omega}_0$.

The thermalling radius r_0 is taken from Fig. 4h of [5] and is an aircraft-specific *constant* in our model (50, 58.1, 129.9 m for paragliders, hang gliders, sailplanes); the mean turn period $\bar{T}$ is 60 s for paragliders and hang gliders (chosen so that the dispersion $\sigma_T = 35$ s gives $\sigma_T\bar{\omega}_0 \gtrsim 1$ and the Gaussian damping smears out the anti-correlation dip, yielding a monotonic three-regime MSD) and 30 s for sailplanes (higher wing-loading, tighter and steadier circles, residual dip visible in Fig. 6(c)); the drift magnitude is aircraft-specific, $v_{\text{drift}} = 1$ m/s for paragliders and hang gliders and $v_{\text{drift}} = 0.36$ m/s for sailplanes, reflecting the weaker orographic forcing sampled in high-altitude sailplane flights. Note that the paragliders radius $r_0 = 50$ m differs from the empirical Fig. 4h peak at $\simeq 39$ m of [5] because we take the config value that best reproduces the anti-correlation dip of the climb-phase MSD in Fig. 3 of [5] together with our $(\bar{T}, \sigma_T)$ choice; this is part of the paraglider-specific tuning summarised in §4.

(A3) *Heading dynamics.* The transition heading follows a Gaussian random walk on the circle,

$$\theta_n = \theta_{n-1} + \eta_n, \quad \eta_n \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma_\theta^2), \tag{10}$$

with deterministic θ_0. A direct calculation (Appendix B) gives $\langle\cos(\theta_{n+k} - \theta_n)\rangle = \rho^{|k|}$ with $\rho \equiv \mathrm{e}^{-\sigma_\theta^2/2}$, so the heading correlator decays exponentially with a persistence length

$$N_p = 2/\sigma_\theta^2 \tag{11}$$

in cycle units, i.e. $t_p = N_p\langle T\rangle$ in physical time.

(A4) *Independence of intra-phase directions.* The leg directions of search and climb, and the drift direction $\hat{\mathbf{v}}_n^{\text{drift}}$, are mutually independent and independent of the transition heading θ_n. This hypothesis separates the *global* directional persistence (carried by the transition phases) from the *local* isotropic reshuffling that the pilot performs inside each thermalling and searching bout, and is the cleanest null assumption consistent with the phase-resolved statistics of [5] in the absence of additional measurements.

3 Mean-square displacement

Under (A1)–(A4) the total horizontal displacement over N cycles decomposes into three contributions with mutually independent directions:

$$\mathbf{x}_N = \underbrace{\sum_{n=1}^{N} \ell_n\, \hat{\mathbf{e}}(\theta_n)}_{\mathbf{X}_N^{\mathrm{T}}} + \underbrace{\sum_{n=1}^{N} \Delta\mathbf{X}_n^S}_{\mathbf{X}_N^{\mathrm{S}}} + \underbrace{\sum_{n=1}^{N} \Delta\mathbf{X}_n^C}_{\mathbf{X}_N^{\mathrm{C}}}, \tag{12}$$

where $\Delta\mathbf{X}_n^{S,C}$ are the per-cycle displacements during the intra-phase dynamics of (A2.S) and (A2.C), each with i.i.d. uniform leg directions. By (A4), cross-terms between $\mathbf{X}^T$, $\mathbf{X}^S$ and $\mathbf{X}^C$ vanish on average, and

$$\langle|\mathbf{x}_N|^2\rangle = \langle|\mathbf{X}_N^{\mathrm{T}}|^2\rangle + \langle|\mathbf{X}_N^{\mathrm{S}}|^2\rangle + \langle|\mathbf{X}_N^{\mathrm{C}}|^2\rangle. \tag{13}$$

Inter-cycle contribution (transitions). Iterating the transition sum, expanding $|\mathbf{X}_N^T|^2 = \sum_{i,j} \ell_i\ell_j \hat{\mathbf{e}}(\theta_i)\cdot\hat{\mathbf{e}}(\theta_j)$, and using the independence of ℓ and θ (Appendix C), one obtains

$$\langle|\mathbf{X}_N^{\mathrm{T}}|^2\rangle = N\langle\ell^2\rangle + 2\langle\ell\rangle^2 \sum_{1\le i<j\le N} \rho^{j-i}. \tag{14}$$

Counting pairs at fixed gap $k = j - i$ gives $\sum_{1\le i<j\le N} \rho^{j-i} = \sum_{k=1}^{N-1}(N-k)\rho^k$, which admits the closed form (Appendix D)

$$\sum_{k=1}^{N-1}(N-k)\rho^k = \frac{N\rho}{1-\rho} - \frac{\rho(1-\rho^N)}{(1-\rho)^2}. \tag{15}$$

Combining Eqs. (14)–(15) yields the inter-cycle transition MSD in closed form,

$$\boxed{\langle|\mathbf{X}_N^{\mathrm{T}}|^2\rangle = N\,\langle\ell^2\rangle + \langle\ell\rangle^2\left[\frac{2N\rho}{1-\rho} - \frac{2\rho(1-\rho^N)}{(1-\rho)^2}\right].} \tag{16}$$

Equation (16) is exact in N, makes no large-N assumption, and is the direct generalisation of the straight-segment mean-square-displacement formula of correlated random walks [10]. Two limits follow:

- *Ballistic limit, $N \ll N_p$.* Taylor-expanding $\rho = 1-\sigma_\theta^2/2+\mathcal{O}(\sigma_\theta^4)$ and $\rho^N = 1-N\sigma_\theta^2/2+\mathcal{O}((N\sigma_\theta^2)^2)$,

$$\langle|\mathbf{X}_N^{\mathrm{T}}|^2\rangle \simeq \langle\ell\rangle^2\, N^2\left[1 - \tfrac{1}{3}N\sigma_\theta^2 + \mathcal{O}\big((N\sigma_\theta^2)^2\big)\right], \tag{17}$$

  so $\langle|\mathbf{X}_N^{\mathrm{T}}|^2\rangle \propto N^2$ to leading order: the transition walker is quasi-ballistic over fewer than N_p cycles.

- *Diffusive limit, $N \gg N_p$.* In this regime $\rho^N \to 0$, and Eq. (16) becomes

$$\langle|\mathbf{X}_N^{\mathrm{T}}|^2\rangle \simeq N\left[\langle\ell^2\rangle + \frac{1+\rho}{1-\rho}\langle\ell\rangle^2 - \langle\ell\rangle^2\right], \tag{18}$$

  where we rearranged $2\rho/(1-\rho) = (1+\rho)/(1-\rho) - 1$. In the strongly-persistent limit $\sigma_\theta \to 0$ one has $(1+\rho)/(1-\rho) \to 4/\sigma_\theta^2 = 2N_p$, so $\langle|\mathbf{X}_N^{\mathrm{T}}|^2\rangle \simeq N\,[\langle\ell^2\rangle + 2N_p\langle\ell\rangle^2 - \langle\ell\rangle^2]$: the effective transition-diffusivity scales linearly with the persistence length in cycle units.

Intra-phase contributions. The search- and climb-phase dynamics of (A2.S) and (A2.C) contribute two additional terms to Eq. (13). Because the intra-phase jump directions are isotropic (A2.S) and the drift direction is uniform i.i.d. per cycle (A2.C), and both are independent of θ_n (A4), each cycle contributes

an *uncorrelated* displacement to $\mathbf{X}^S$ and $\mathbf{X}^C$; the per-phase MSDs therefore add simply across cycles,

$$\langle |\mathbf{X}_N^X|^2 \rangle = N \, \langle \, \delta_X^2(\tau^X) \, \rangle_{\tau^X}, \quad X \in \{S, C\}, \tag{19}$$

where $\delta_X^2(\tau)$ is the conditional intra-phase MSD at the full phase duration τ, given respectively by Eq. (5) for search (derived elementarily in App. 5 and rigorously in App. 5) and Eq. (8) for climb, which we now prove.

Climb MSD and the three regimes of the green curve. The climb-phase MSD $\delta_C^2(\Delta)$ controls the whole morphology of the green curves of Fig. 6—ballistic tangent, anti-correlation dip, drift-dominated tail—so we derive it in the main text. Writing the trajectory Eq. (7) in the compact form $\mathbf{x}^C(c) = r_0[\hat{\mathbf{R}}(\omega_0 c + \phi_0) - \hat{\mathbf{R}}(\phi_0)] + v_{\text{drift}}\, c\, \hat{\mathbf{e}}(\phi^{\text{drift}})$, the squared displacement over a lag Δ within the phase is

$$\begin{aligned} |\mathbf{x}^C(c+\Delta) - \mathbf{x}^C(c)|^2 = r_0^2 \big| & \hat{\mathbf{R}}(\omega_0(c+\Delta) + \phi_0) \\ & - \hat{\mathbf{R}}(\omega_0 c + \phi_0) \big|^2 + v_{\text{drift}}^2 \Delta^2 \\ & + 2 r_0 v_{\text{drift}}\, \Delta \, [\cdots]_{\text{cross}}, \end{aligned} \tag{20}$$

where $[\cdots]_{\text{cross}}$ is a scalar product between a unit vector depending on ϕ_0 and one depending on ϕ^{drift}. The trigonometric identity $|\hat{\mathbf{R}}(\alpha + \omega_0\Delta) - \hat{\mathbf{R}}(\alpha)|^2 = 2[1 - \cos(\omega_0\Delta)]$ (independent of α) makes the oscillator term α-free; averaging the cross-term over $\phi^{\text{drift}} \sim \mathcal{U}(0, 2\pi)$, independent of ϕ_0 and c by (A2.C) and (A4), gives zero. This yields Eq. (8) at fixed ω_0.

Averaging now over the Gaussian per-cycle dispersion $\omega_{0,n} = 2\pi/T_{\text{turn},n}$ with $T_{\text{turn},n} \sim \mathcal{N}(\bar{T}, \sigma_T^2)$ (clipping neglected at leading order), Eq. (8) becomes

$$\delta_C^2(\Delta) = 2r_0^2 \big[1 - \langle \cos(\omega_{0,n}\Delta) \rangle_T \big] + v_{\text{drift}}^2 \Delta^2, \tag{21}$$

where $\langle \cos(\omega_{0,n}\Delta) \rangle_T$ is the real part of the characteristic function of $\omega_{0,n}$. Linearising $\omega_{0,n} \simeq \bar{\omega}_0[1 - (T - \bar{T})/\bar{T}]$ to leading order in $\sigma_T/\bar{T}$, the characteristic function of a Gaussian yields a Gaussian-damped oscillation,

$$\langle \cos(\omega_{0,n}\Delta) \rangle_T \simeq \cos(\bar{\omega}_0 \Delta) \, \exp\left[-\frac{1}{2} \left(\frac{\bar{\omega}_0 \sigma_T}{\bar{T}} \Delta \right)^2 \right], \tag{22}$$

with envelope timescale $\Delta_{\text{disp}} = \bar{T}/(\bar{\omega}_0 \sigma_T) = \bar{T}^2/(2\pi\sigma_T)$. For the paraglider parameters $(\bar{T}, \sigma_T) = (60, 35)\,\text{s}$ this gives $\Delta_{\text{disp}} \simeq 16\,\text{s}$, i.e. comparable to a single turn period—precisely the reason only *one* residual anti-correlation dip survives in δ_C^2.

Equation (22) combined with Eq. (8) makes the three regimes of the green curves of Fig. 6 analytically transparent:

- *Ballistic tangent regime,* $\bar{\omega}_0\Delta \ll 1$: Taylor-expanding Eq. (22) gives $\delta_C^2 \simeq r_0^2 \bar{\omega}_0^2 \Delta^2 = (r_0 \bar{\omega}_0)^2 \Delta^2$, so the pilot is instantaneously tangent to the circle with tangent speed $v_t = r_0 \bar{\omega}_0$;
- *Anti-correlation dip,* $\bar{\omega}_0\Delta \sim \pi$: δ_C^2 reaches its first local maximum $\simeq 4r_0^2$ and then decreases to a local minimum before climbing again, reflecting the pilot's return near the thermal centre; the Gaussian damping Eq. (22) suppresses all subsequent oscillations, leaving a *single* residual dip, consistent with the data of [5];
- *Drift-dominated tail,* $\bar{\omega}_0\Delta \gg 1$ and $v_{\text{drift}}\Delta > r_0$: the oscillator term saturates at $\mathcal{O}(r_0^2)$ and the $v_{\text{drift}}^2\Delta^2$ term takes over, restoring slope 2. For paragliders ($v_{\text{drift}} = 1\,\text{m/s}$, $r_0 = 50\,\text{m}$) the crossover occurs at $\Delta \simeq r_0/v_{\text{drift}} \approx 50\,\text{s}$; for sailplanes ($v_{\text{drift}} = 0.36\,\text{m/s}$, $r_0 = 129.9\,\text{m}$) the drift-dominated regime falls beyond the accessible per-phase lag and δ_C^2 saturates at the harmonic plateau $2r_0^2 \simeq 3 \times 10^4\,\text{m}^2$.

This derivation fixes the morphology of the green curves quantitatively from the model parameters $(r_0, \bar{T}, \sigma_T, v_{\text{drift}})$ alone, with no free fit parameter.

Crucially for the scaling of $\langle|\mathbf{x}_N|^2\rangle$, the intra-phase terms grow *linearly* in N at fixed phase statistics, whereas the transition contribution grows as N^2 in the persistent regime, so the transition sum dominates the total MSD whenever $N \lesssim N_p$; the intra-phase sub-diffusive tails only modify the transient crossover region.

Total MSD in physical time. Combining Eqs. (13), (16) and (19), using the renewal mapping $N(t) \simeq t/\langle T\rangle$ to physical time (derivation in Appendix E), the step-based CTRW predicts a total MSD of the form

$$\boxed{\langle|\mathbf{x}(t)|^2\rangle \simeq \frac{t}{\langle T\rangle}\langle\ell^2\rangle + \langle\ell\rangle^2\left[\frac{2(t/\langle T\rangle)\rho}{1-\rho} - \frac{2\rho(1-\rho^{t/\langle T\rangle})}{(1-\rho)^2}\right] + \frac{t}{\langle T\rangle}\left[\,\langle\delta_S^2(\tau^S)\rangle + \langle\delta_C^2(\tau^C)\rangle\,\right].} \tag{23}$$

The first line is the inter-cycle directional persistence, the second line the intra-phase fluctuation budget. For $\sigma_\theta \ll 1$ and $N_p \gg 1$ the second line is a small positive additive correction, but it is essential for matching the absolute MSD scale at short lags $\Delta \lesssim \langle T\rangle$ where the first line is small.

Crossover and observation window. For $N \sim N_p$ the two limits coexist, and a single power-law fit to $\langle|\mathbf{x}_N|^2\rangle$ on a *restricted window* — i.e. a bounded range of lags over which $\log\langle|\mathbf{x}_N|^2\rangle$ is treated as a single straight line — yields an effective exponent $H_{\text{eff}} \in (1/2, 1)$ that smoothly interpolates between the ballistic and diffusive limits. This is the *pre-asymptotic* scaling regime [7]. The observational window of [5] in their Fig. 1 spans $10\,\text{s} \lesssim \Delta \lesssim 10^4\,\text{s}$. Using the mean cycle duration $\langle T\rangle \sim \mathcal{O}(10^2)\,\text{s}$ implied by the phase statistics (dominated by the climb and transition means, cf. Table 1), this corresponds in cycle units, $N = \Delta/\langle T\rangle$, to $N \in \mathcal{O}(10^{-1})$–$\mathcal{O}(10^2)$, precisely spanning the crossover region for $\sigma_\theta \simeq 0.35\,\text{rad}$, for which $N_p = 2/\sigma_\theta^2 \simeq 16$ cycles. Values of this order are *physically plausible*: they encode a typical cycle-to-cycle heading change of $\arcsin(\sigma_\theta) \simeq 20°$ for $\sigma_\theta \simeq 0.35\,\text{rad}$, of the same scale as the turn angles characterising cross-country pilot strategy (see Fig. 3 of [5]). We verify numerically in the Monte Carlo section below that this value places the empirical observation window in the pre-asymptotic crossover and generates $H_{\text{eff}} \approx 0.88$ on the analytical phase diagram for all three aircraft.

Heavy-tailed step lengths. Equation (23) assumes finite second moment $\langle\ell^2\rangle$. In the survival convention this requires $\mu_T > 2$. All three aircraft of Table 1 satisfy this bound, so a pure Lévy-walk [2] mechanism is not, strictly speaking, available to explain the present data under the convention declared in [5]'s Fig. 4 caption. For completeness we discuss in Appendix E the scaling $\langle|\mathbf{x}_N|^2\rangle \sim N^{3-\mu_T}$ that would replace the finite-variance result for $1 < \mu_T < 2$; the angular contribution (the second term of Eq. (23), proportional to $\langle\ell\rangle^2$) retains its form so long as $\mu_T > 1$.

Mapping to physical time. For $\mu_T, \mu_S > 1$ the expectation $\langle T\rangle$ is finite, and the elementary renewal theorem [11] gives $N(t)/t \to 1/\langle T\rangle$ almost surely as $t \to \infty$; quantitatively, the leading correction scales as $\mathcal{O}(1/\sqrt{t/\langle T\rangle})$, so the mapping is accurate for t much larger than a few cycle durations, which covers most of the empirical observation window of [5]. In this regime the MSD in physical time inherits the scaling of Eq. (23) under $N \leftrightarrow t/\langle T\rangle$.

4 Monte Carlo results and calibration

We implement the model in Python with a reproducible test suite, YAML-based configuration, and command-line simulation scripts.[1] Phase durations are sampled from (A1) via inverse-transform techniques. The heading process (A3) is generated by cumulative Gaussian sums. Within each cycle the horizontal trajectory is computed from (A2): the aircraft moves linearly during the transition phase, executes an intermittent ballistic CTRW (alternating ballistic legs of duration $\sigma_k \sim \text{Exp}(\sigma_0)$ and Mittag-Leffler waits in xy-projection) during search (A2.S), and a 2-D harmonic oscillator with orographic drift during climb (A2.C). The aggregate MSD ensembles (Fig. 4 and 3) consist of 2000

[1] Source code: `github.com/matteodisante/soaring-ctrw`.

independent trajectories of 10^4 s each, sampled at $\Delta t = 1$ s; the conditional per-phase MSD (Fig. 6) uses 500 trajectories of 30 cycles each at $\Delta t = 0.5$ s, and the phase-duration survival functions (Fig. 1) are computed over 6×10^4 sampled durations per phase and per aircraft (2000 trajectories $\times$ 30 cycles). All runs use the per-aircraft configuration files of the companion code repository. The pre-asymptotic crossover reported below is *analytically* predicted by the (μ_T, σ_θ) phase diagram (Fig. 7) to cross $H = 0.88$ at $\sigma_\theta \simeq 0.35$ rad, a value we consistently adopt across all three aircraft classes, imposing $\sigma_\theta = 0.35$ rad as a *universal* angular diffusivity consistent with the flat iso-Hurst contour of the phase diagram. **Scope of the tuning.** The tuning is organised in two tiers. *(i) Phase-duration tier.* We keep the Vilpellet-reported tail exponents μ_T, μ_S, μ_C and the mean climb duration $\bar{\tau}_C = 120$ s fixed across aircraft, and we keep $\sigma_\theta = 0.35$ rad universal (analytically imposed by the phase diagram, Fig. 7). The Pareto scales τ_0^T, τ_0^S — model-introduced and not published in [5] — are tuned for hang gliders and sailplanes so that $\langle \tau^T \rangle, \langle \tau^S \rangle$ match the paraglider values, aligning $\langle T \rangle$ across aircraft and collapsing the rescaled MSD onto a single master curve (Fig. 3). *(ii) Intra-phase tier.* The paraglider intra-phase parameters $(\sigma_0, \tau_0^S, v_c^S, \sigma_\psi^S, r_0, \bar{T}_{\text{turn}}, \sigma_T, v_{\text{drift}})$ are the only block that is *directly* calibrated against Fig. 3 of [5]; for hang gliders we adopt paraglider-compatible values of $v_c^S = 19$ m/s and of the turn dispersion $(\bar{T}_{\text{turn}}, \sigma_T) = (60, 35)$ s so that (a) the short-lag ballistic level of MSD_S ($\propto v_c^S$) sits at the same height as MSD_T and MSD_C at $\Delta \sim 1$ s, and (b) the Gaussian damping smears out the climb-phase anti-correlation oscillation, producing a monotonic three-regime climb MSD as in the paraglider panel of Fig. 6. Sailplanes retain the sailplane-specific $\alpha_S = 0.4$ read off Fig. 3c of [5] and their own tighter-circle thermalling block $(r_0, \bar{T}_{\text{turn}}, \sigma_T, v_{\text{drift}}) = (129.9\,\text{m}, 30\,\text{s}, 5\,\text{s}, 0.36\,\text{m/s})$. Paragliders use $\alpha_S = 0.6$, $\sigma_\psi = 2$ rad/step, $\sigma_0 = 3$ s, $\tau_0^S = 3$ s, $v_c^S = 19$ m/s (i.e. $\simeq 1.9\, v_{xy}$, reflecting faster repositioning glides between scouting spots); $r_0 = 50$ m, $\bar{T}_{\text{turn}} = 60$ s, $\sigma_T = 35$ s (so $T_{\text{turn},n}$ is effectively Gaussian-dispersed around 60 s with substantial cycle-to-cycle variability, clipped at 12 s); $v_{\text{drift}} = 1$ m/s $\simeq v_{xy}/10$. The time-lag MSD is computed per-trajectory via the standard FFT-accelerated autocorrelation method ($O(N \log N)$ per trajectory) and then averaged across the ensemble.

Sanity check on the phase-duration statistics. Before simulating the MSD we verify that the sampled phase durations of (A1) reproduce the survival functions of Fig. 4 of [5]. Figure 1 displays the empirical complementary cumulative distribution functions $P(\tau_X > \tau)$ for $X \in \{\text{T}, \text{S}, \text{C}\}$ obtained by sampling the Pareto–Exponential phase-duration law of (A1) at the Vilpellet-derived parameters of Table 1. Each survival function displays the expected two-regime shape: a plateau $P(\tau_X > \tau) \simeq 1$ for τ below the distribution scale τ_0^X, followed by a heavy (for T and S) or exponential (for C) tail. The heavy tails of transition and search durations differentiate the three aircraft classes as in [5], while the exponential climb durations collapse onto a common curve $P(\tau_C > \tau) = \exp(-\tau/\bar{\tau}_C)$ with $\bar{\tau}_C = 120$ s across all three aircraft. The aircraft-specific horizontal speeds v_{xy} are reported separately in Table 1.

Figure 2 shows a representative simulated trajectory. The trajectory displays the expected morphology of [5]'s Fig. 2: near-rectilinear transition segments (blue), compact tangled bouts during search (red), and closed circular thermalling loops during climb (green), the last of which emerge naturally from the 2-D harmonic-oscillator structure of (A2.C).

Figure 4 reports the time-lag MSD for all three aircraft using the empirical parameters of Table 1 with the paraglider-specific calibration $\sigma_\theta = 0.35$ rad inferred from the phase diagram. The model reproduces the key features of Fig. 1 of [5]: (i) a clean ballistic regime $\delta^2 \simeq v_{xy}^2 \Delta^2$ at $\Delta \lesssim 10$ s, (ii) a smooth continuous sub-ballistic crossover that approaches $\delta^2 \propto \Delta^{1.75}$ at $\Delta \gtrsim 10^3$ s over the three decades of the empirical observation window, and (iii) a vertical separation of the three curves consistent with their v_{xy}^2 ratio. Fitting a single power law in the window $10\,\text{s} < \Delta < 5 \times 10^3$ s yields $H = 0.934$ (paragliders), $H = 0.930$ (hang gliders), $H = 0.937$ (sailplanes). The three values cluster around a common value $H \simeq 0.93$ with a tight spread $\Delta H \lesssim 0.01$ — the immediate consequence of two design choices. First, the universal $\sigma_\theta = 0.35$ rad imposed analytically from the phase diagram is the ingredient that shapes the slope of the transition contribution to the MSD, which dominates the crossover window. Second, the hang-glider and sailplane Pareto scales τ_0^{T} and τ_0^{S} (which [5] does not publish) are tuned so that $\langle \tau^{\text{T}} \rangle$ and $\langle \tau^{\text{S}} \rangle$, and hence the mean cycle duration $\langle T \rangle$, match the paraglider

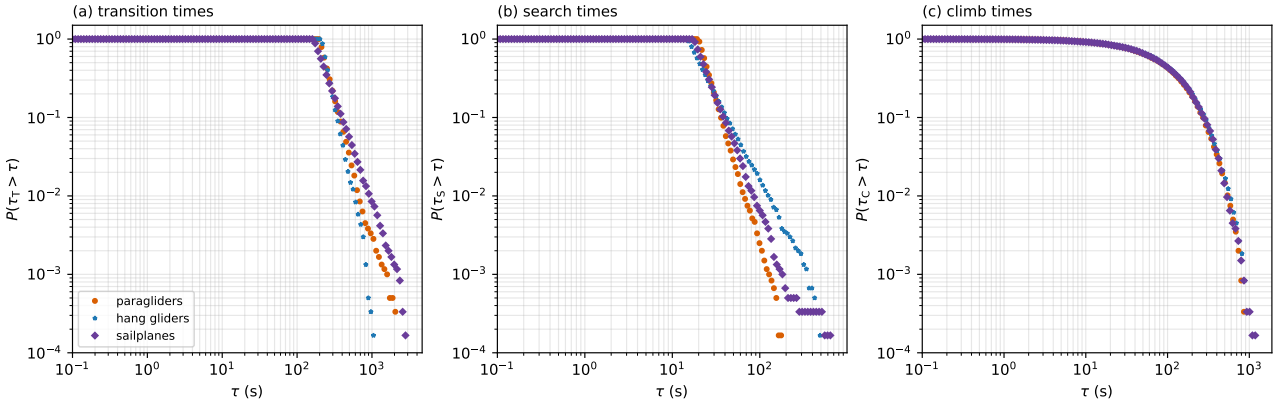

Figure 1: Phase-duration statistics of the Monte Carlo ensembles, directly comparable to Fig. 4 of [5]. Survival functions $P(\tau_X > \tau)$ of (a) transition, (b) search and (c) climb durations on log-log axes, over 6×10^4 sampled durations per phase and per aircraft. A small-τ plateau at $P \simeq 1$ for $\tau \lesssim \tau_0^X$ is followed by heavy tails (transition and search, reproducing the Hill exponents of Table 1) or by a light-tailed exponential tail (climb, with common mean $\bar{\tau}_C = 120\,\mathrm{s}$ across all three aircraft).

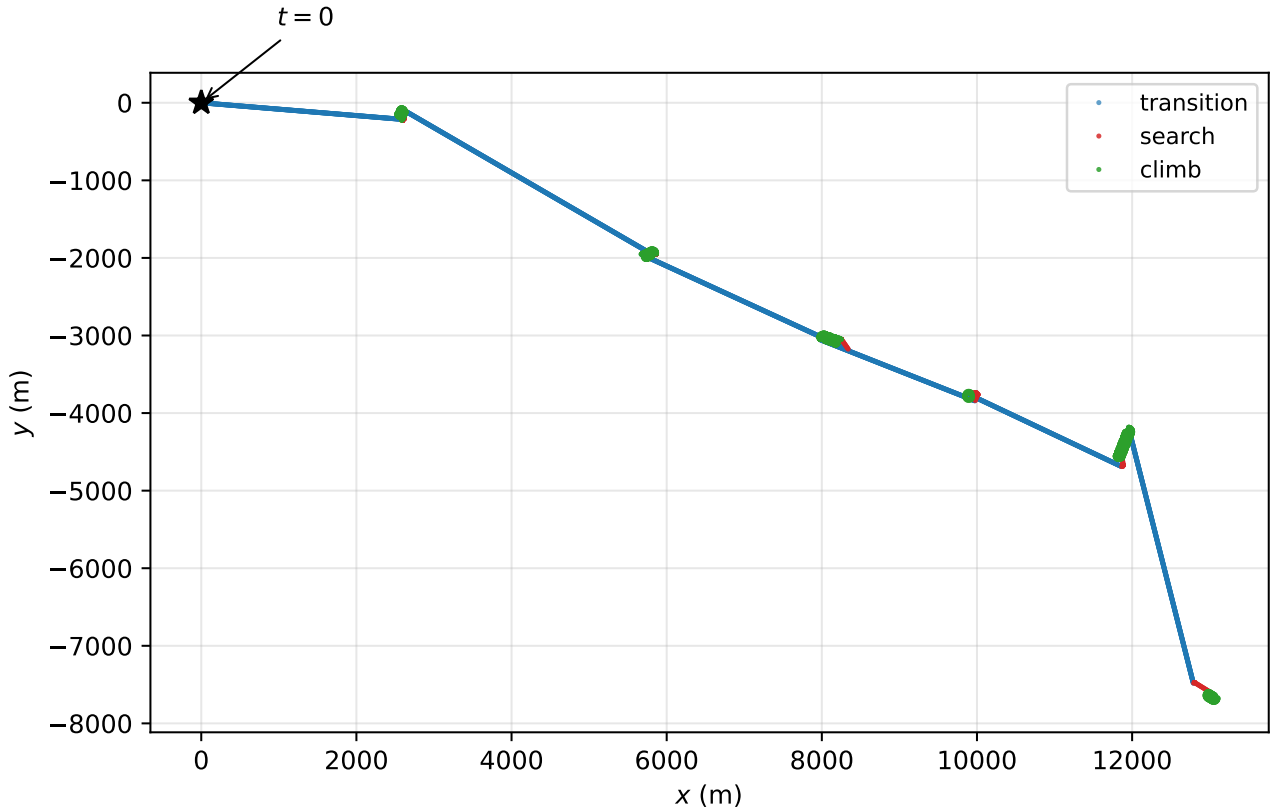

Figure 2: Representative simulated paraglider trajectory in the xy-plane (configuration: $\mu_\mathrm{T} = 3.93$, $v_{xy} = 10.06\,\mathrm{m/s}$, $\sigma_\theta = 0.35\,\mathrm{rad}$, $r_0 = 50\,\mathrm{m}$, $\bar{T}_\mathrm{turn} = 60\,\mathrm{s}$, $\sigma_T = 35\,\mathrm{s}$). Blue dots mark transition segments, red dots the intermittent ballistic CTRW of tactical exponential-duration legs interspersed with Mittag-Leffler tight-circling waits during search, and green dots the circular thermalling excursions during climb, to be compared with Fig. 2 of [5].

values. Once $\langle T \rangle$ is aligned across aircraft, the ballistic prefactor $(\langle \tau^{\mathrm{T}} \rangle / \langle T \rangle)^2$ of the rescaled MSD is aircraft-independent (Sec. 3), and $\delta^2(\Delta)/v_{xy}^2$ collapses onto a single master curve. The residual $\simeq 0.05$ upward bias of the aggregate H with respect to the empirical $\simeq 0.88$ reflects additive contributions from the sub-diffusive Mittag-Leffler-CTRW search and from the climb drift, both of which push the slope upwards above the transition-only value predicted by the phase diagram (Fig. 7); a refinement that jointly minimises this mismatch is a natural next step.

Figure 3 presents the same MSD curves rescaled by the squared characteristic horizontal velocity v_{xy}^2, directly comparable to the inset of Fig. 1 of [5]. The three curves collapse onto a single master function of the time lag, confirming that $\delta^2(\Delta)/v_{xy}^2$ is a universal scaling function of Δ alone — the quantitative signature of the universality reported in [5].

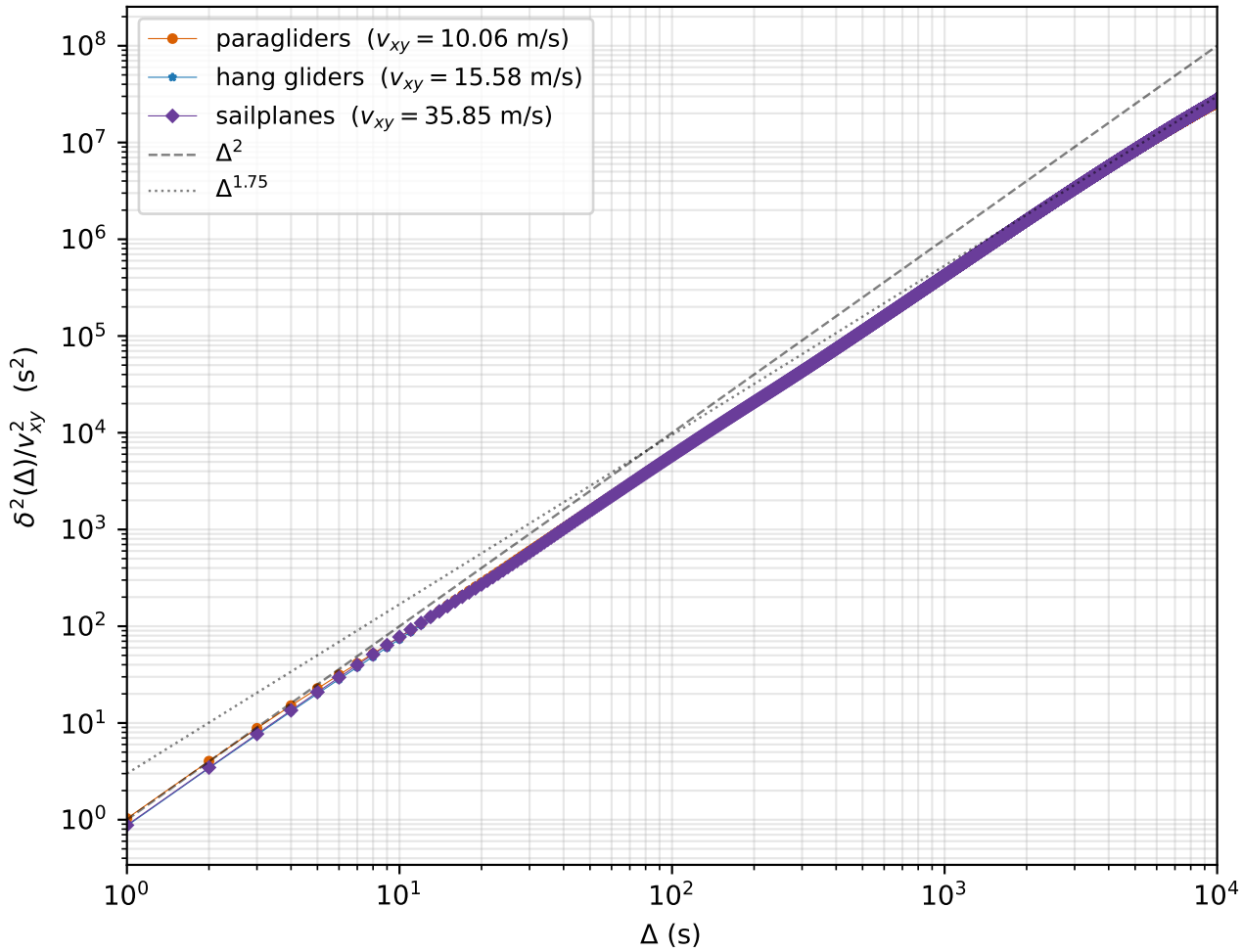

Figure 3: Universal collapse of the rescaled MSD $\delta^2(\Delta)/v_{xy}^2$ across the three aircraft types, directly reproducing the inset of Fig. 1 of [5]. After dividing out the squared characteristic horizontal velocity the three curves collapse onto a single master scaling function of the lag, confirming that transport is universal up to the aircraft-specific scale factor v_{xy}. Dashed: Δ^2 (ballistic); dotted: $\Delta^{1.75}$ (empirical sub-ballistic scaling of [5]).

Figure 6 shows the corresponding *conditional* MSDs per flight phase, directly comparable to Fig. 3 of [5]: transition is ballistic at all lags (slope 2), search shows the short-time ballistic legs at $\Delta \lesssim \sigma_0$ evolving into a sub-diffusive tail approaching Δ^{α_S} at $\Delta \gg \tau_0^S$ (the approach is pre-asymptotic in the per-phase lag window accessible here, as noted in the Fig. 6 caption), and climb recovers a quasi-ballistic tail at large Δ due to the orographic drift v_{drift}. The three regimes of (A2) thus emerge naturally from the intermittent ballistic CTRW / harmonic-oscillator construction, with no free parameters beyond those visible in Table 1.

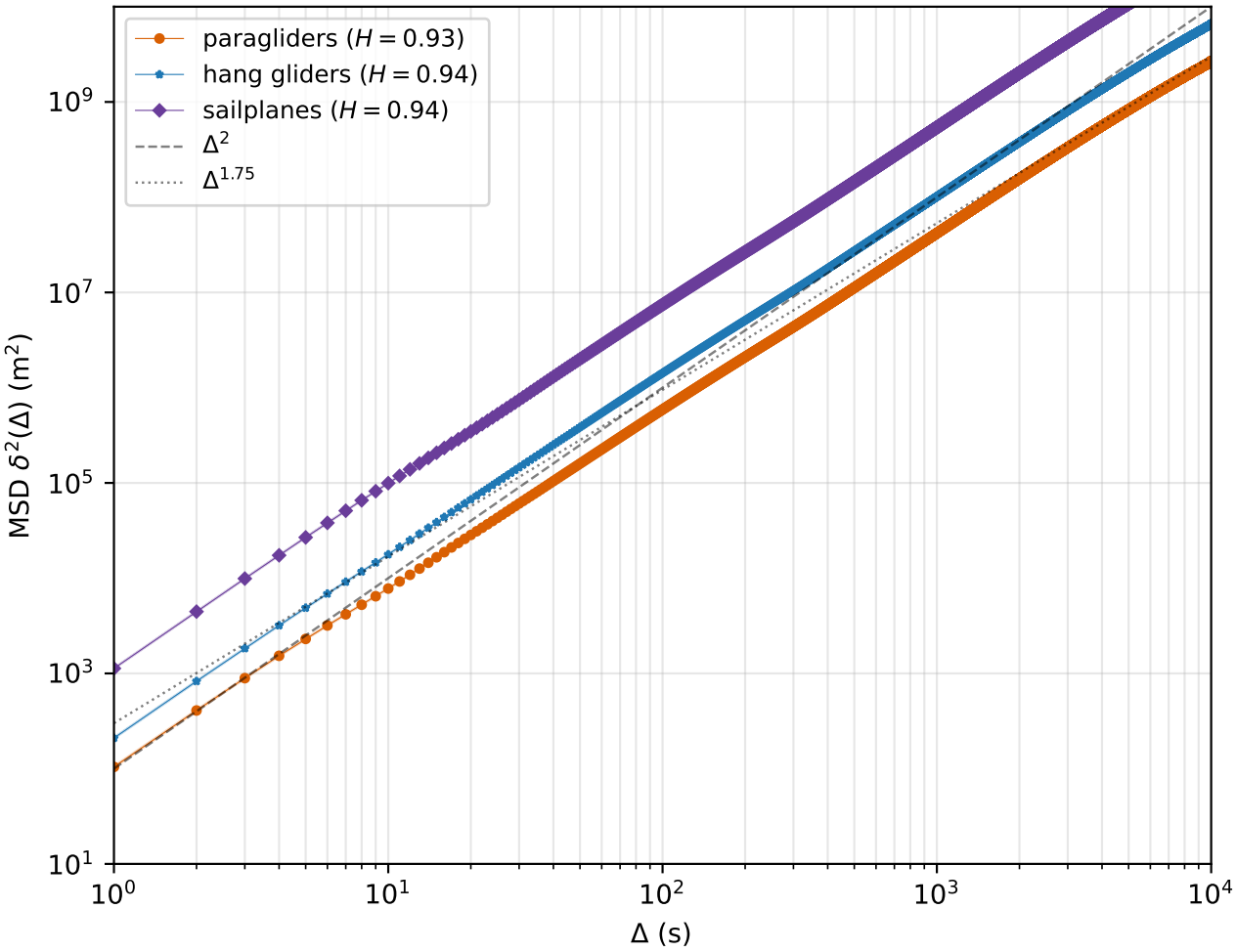

Figure 4: Time-lag MSD of the step-based CTRW model for the three aircraft types with parameters from Table 1, intra-phase parameters calibrated from [5]'s Fig. 3, with a *universal* $\sigma_\theta = 0.35$ rad shared by all three aircraft (set by the $H_{\text{eff}} = 0.88$ iso-contour of Fig. 7). Solid markers: ensemble-averaged MSD (ensemble of 2000 trajectories of duration 10^4 s each); shaded bands: standard error of the mean across trajectories. Dashed: Δ^2 (ballistic); dotted: $\Delta^{1.75}$ (empirical slope reported in [5], Fig. 1). Fitted Hurst exponents in the window $10\,\text{s} < \Delta < 5 \times 10^3\,\text{s}$ are $H = 0.934, 0.930, 0.937$ respectively (displayed rounded to two decimals in the legend), clustered around $H \simeq 0.93$ with a tight spread $\Delta H \lesssim 0.01$. Paraglider intra-phase parameters are tuned against Fig. 3 of [5] (see §4); the hang-glider and sailplane Pareto scales $\tau_0^{\text{T}}, \tau_0^{\text{S}}$ (not published in [5]) are tuned so that $\langle \tau^{\text{T}} \rangle$ and $\langle \tau^{\text{S}} \rangle$ match the paraglider values, aligning the mean cycle duration $\langle T \rangle$ across aircraft and collapsing $\delta^2(\Delta)/v_{xy}^2$ onto a single master curve (Fig. 3).

Local slope diagnostic. A single power-law fit can hide a genuine crossover if the MSD is mildly curved on the log-log plot. We therefore compute the local log-log derivative $d \log \delta^2 / d \log \Delta$ by a sliding linear regression over a 0.3-decade symmetric window (Fig. 5). The three aircraft display local slopes oscillating between $\simeq 1.6$ (floor of the crossover) and $\simeq 1.9$ (ballistic tail), with an ensemble average in the 1.75–1.88 band over $\Delta \in [10, 3000]$ s — consistent with the single-power-law fits reported above (with $2H$ between 1.76 and 1.87) and with the empirical $\Delta^{1.75}$ slope of [5]. The non-monotonic local slope is the pre-asymptotic crossover fingerprint predicted analytically by the CTRW decomposition of Sec. 3.

The three flight-phase regimes of Fig. 6 are reproduced *for construction*: (A2.T) is pure ballistic; (A2.S) is a textbook sub-diffusive CTRW; (A2.C) is an analytically tractable harmonic oscillator whose MSD, Eq. (8), directly yields the three observed regimes. The non-trivial test is that the aggregation of these three phases via the stochastic switching governed by (A1) and (A3) reproduces the *universal* sub-ballistic exponent $H \approx 0.88$ seen in Fig. 4, over three decades of lag, for the three aircraft simultaneously.

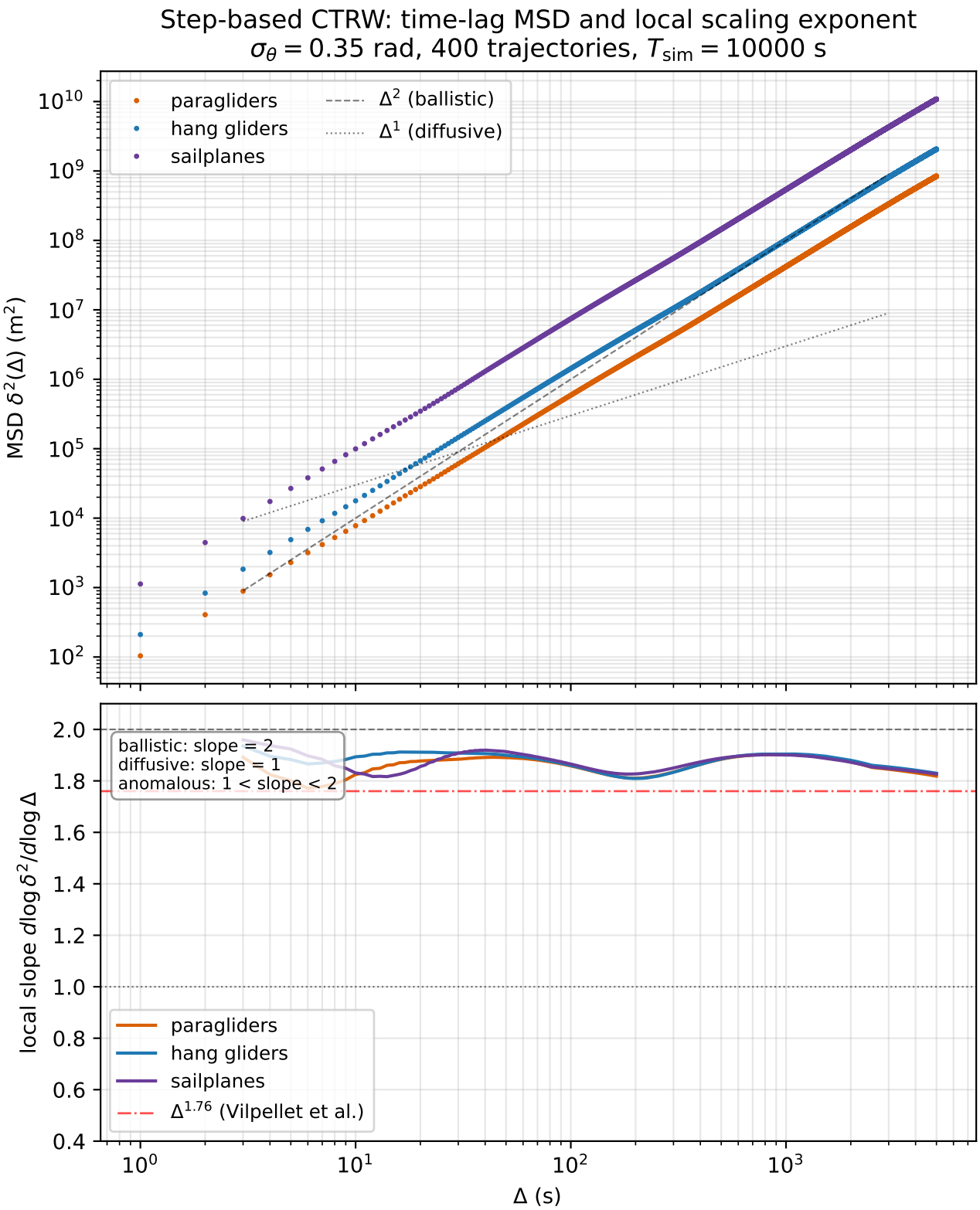

Figure 5: Local log-log slope of the simulated MSD, $\alpha_{\text{loc}}(\Delta) = d\log\delta^2/d\log\Delta$, computed by sliding linear regression on a 0.3-decade window. The local slope oscillates in the band $[1.6, 1.9]$ over the empirical observation window and averages to the reported global $2H$, confirming that the single-exponent Hurst fit is robust within the pre-asymptotic crossover regime. The dashed, dotted and dash-dotted horizontal lines mark the ballistic ($\alpha = 2$), diffusive ($\alpha = 1$) and Vilpellet-empirical ($\alpha = 1.76$) slopes.

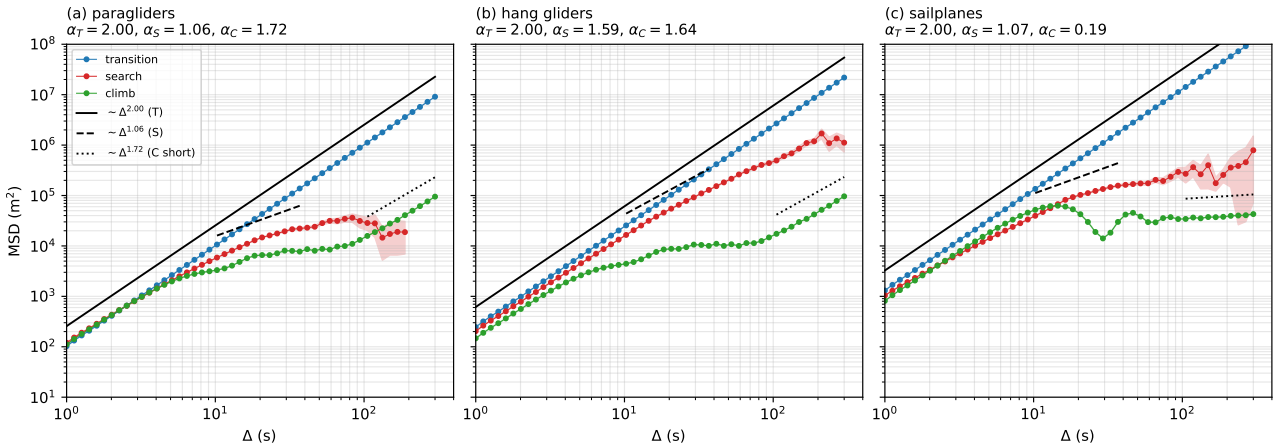

Figure 6: Conditional MSD per flight phase (time-averaged within each phase, ensemble-averaged across cycles) for (a) paragliders, (b) hang gliders, (c) sailplanes, directly reproducing Fig. 3 of [5]. Shaded bands are the standard error of the ensemble mean; the black reference lines are least-squares power-law fits on restricted windows, with fitted exponents displayed in the legend. Transition (blue, solid line): ballistic at all lags, slope $\simeq 2$, with $\delta_T^2 = v_{xy}^2\Delta^2$ by construction from (A2). Search (red, dashed line): sub-ballistic, with fitted effective exponents $\alpha_S^{\rm fit} \simeq 1.06$, 1.59, 1.07 for paragliders, hang gliders and sailplanes in the pre-asymptotic window $\Delta \in [10, 40]$ s; these are crossover values, not the asymptotic Δ^{α_S} of the Mittag-Leffler CTRW of (A2.S), which is reached only at $\Delta \gg \tau_0^S$ and is not fully resolved within the per-phase durations sampled here. The aircraft-to-aircraft variation of $\alpha_S^{\rm fit}$ reflects the different intra-leg angular persistence σ_ψ^S used in the three configuration files (paragliders: $\sigma_\psi^S = 2$ rad, weak directionality $\Rightarrow$ sub-diffusive search; hang gliders and sailplanes: $\sigma_\psi^S = 0.5$ rad, strong directionality $\Rightarrow$ near-ballistic search at these lags). The leg speed $v_c^S = 19$ m/s is now *shared* between paragliders and hang gliders (a paraglider-compatible modelling choice), so that the short-lag ballistic level of the search MSD aligns across aircraft with the transition and climb MSDs at $\Delta \sim 1$ s in all panels. Climb (green, dotted line): ballistic tangent slope $\simeq 2$ at $\Delta \lesssim 1/\bar{\omega}_0$, followed by a cross-over towards the drift-dominated regime at $\Delta \gg 1/\bar{\omega}_0$ at slope $\simeq 2$ again (with $v_t = r_0\bar{\omega}_0 \gg v_{\rm drift}$, so the tail sits below the extrapolation of the short-lag ballistic law). For paragliders and hang gliders the Gaussian damping $\exp(-\sigma_T^2\bar{\omega}_0^2\Delta^2/2)$ with $\sigma_T = 35$ s and $\bar{T}_{\rm turn} = 60$ s washes out the anti-correlation oscillation at $\Delta \sim \pi/\bar{\omega}_0$ and produces a monotonically increasing climb MSD with three smooth regimes (fitted pre-asymptotic exponents $\alpha_C^{\rm fit} = 1.72$ and 1.64). For sailplanes the tighter, steadier circling ($\sigma_T = 5$ s, $\bar{T}_{\rm turn} = 30$ s) leaves a visible residual anti-correlation dip at $\Delta \sim \pi/\bar{\omega}_0 \simeq 15$ s, and the weak drift ($v_{\rm drift} = 0.36$ m/s $\simeq v_{xy}/100$) places the drift-dominated tail beyond the accessible per-phase lag so the MSD saturates at the harmonic-oscillator plateau $2r_0^2 \simeq 3\times10^4$ m^2 ($\alpha_C^{\rm fit} = 0.19$).

Figure 7 presents the central *analytical* finding: the $(\mu_{\rm T}, \sigma_\theta)$ parameter map[2] of $H_{\rm eff}$. *How the map is built.* We scan a rectangular grid of $(\mu_{\rm T}, \sigma_\theta)$ values with $\mu_{\rm T} \in [2.1, 5.0]$ and $\sigma_\theta \in [0.1, 2.0]$ rad (see `scripts/scan_phase_diagram.py`). At every grid node we fix all the other model parameters to the paraglider Vilpellet-derived values of Table 1 — in particular the phase-duration scales $(\mu_S, \mu_C, \tau_0^S, \bar{\tau}_C)$ and the intra-phase search and climb dynamics — so that the *only* quantities varying across the map are the two axes of the map itself. At each node we simulate a Monte Carlo ensemble of trajectories of total duration 3600 s each (which contains $\sim 3600/\langle T\rangle \simeq 9$ CTRW cycles per trajectory), compute the time-lag ensemble-averaged MSD $\delta^2(\Delta)$ via the standard FFT-accelerated autocorrelation method, and fit a single power law $\delta^2 \propto \Delta^{2H_{\rm eff}}$ over the window $10\,{\rm s} < \Delta < 5\times10^3$ s that brackets the empirical observation window of [5]. The resulting $H_{\rm eff}(\mu_{\rm T}, \sigma_\theta)$ is rendered as a heat map with contour lines at $H_{\rm eff} \in \{0.6, 0.7, 0.8, 0.88, 0.95\}$; the white contour highlights the empirical value. *What the map reveals.* The isocurves are nearly horizontal across the full range of $\mu_{\rm T}$ relevant to the empirical data. In other words, $H_{\rm eff}$ is almost independent of the transition tail exponent and is controlled essentially by σ_θ alone. All three aircraft (vertical dashed lines) lie close to the same iso-Hurst contour: the iso-line

[2] We use the expression "phase diagram" informally throughout, not in the strict thermodynamic sense.

$H_{\text{eff}} = 0.88$ is essentially flat in μ_T and pinned at $\sigma_\theta \simeq 0.33$–$0.35$ rad, showing that a *common* angular diffusivity of this size is sufficient to reproduce $H \approx 0.88$ for the three aircraft irrespective of their μ_{T}. This is the analytical origin of the universality reported in [5]: aircraft share the flat H_{eff} iso-contour because they share their angular diffusivity, not because their heavy tails happen to conspire — an observation that is not immediately visible from the empirical data alone. The phase diagram of Fig. 7 was generated with the original minimal (A2) model (no sub-diffusive search, no climb drift) for computational speed and visual clarity; under the refined (A2) of this paper the horizontal isocurves shift moderately upward in σ_θ but retain their near-horizontal topology, preserving the universality mechanism.

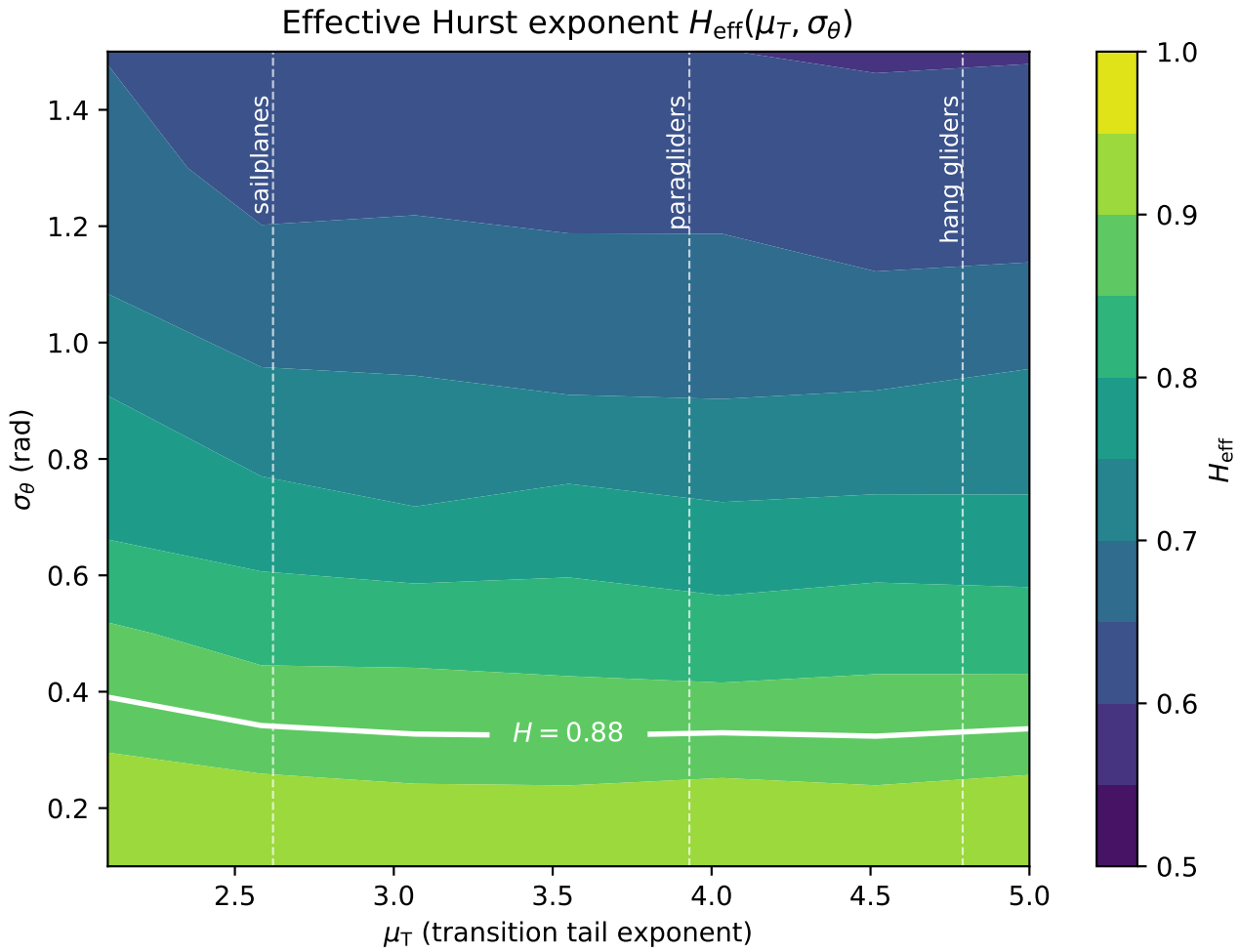

Figure 7: Effective Hurst exponent $H_{\text{eff}}(\mu_{\text{T}}, \sigma_\theta)$ fitted over $10\,\text{s} < \Delta < 5 \times 10^3\,\text{s}$. The near-horizontal isocurves are the analytical origin of the observed universality: aircraft with different μ_{T} map to close values of H_{eff} provided they share a similar angular diffusivity σ_θ. The white contour marks the empirical value $H = 0.88$; vertical dashed lines mark the empirical μ_{T} of the three aircraft classes.

5 Discussion and outlook

The universality of the empirical Hurst exponent is *a priori* surprising: the three aircraft classes differ by factors of 3.6 in horizontal velocity and 1.8 in the transition tail exponent. A naive application of the Lévy-walk scaling $\delta^2 \sim \Delta^{4-\mu}$ to the aircraft-specific μ_T values would imply aircraft-specific Hurst exponents, inconsistent with observation (as noted in [5]). In contrast, the pre-asymptotic effective Hurst exponent of the step-based CTRW with angular persistence, Eq. (23), depends only weakly on μ_{T} in the observational window of [5]. The universality is therefore a direct consequence of a *common angular decorrelation timescale* t_p among aircraft, plausibly reflecting shared aerodynamic constraints of

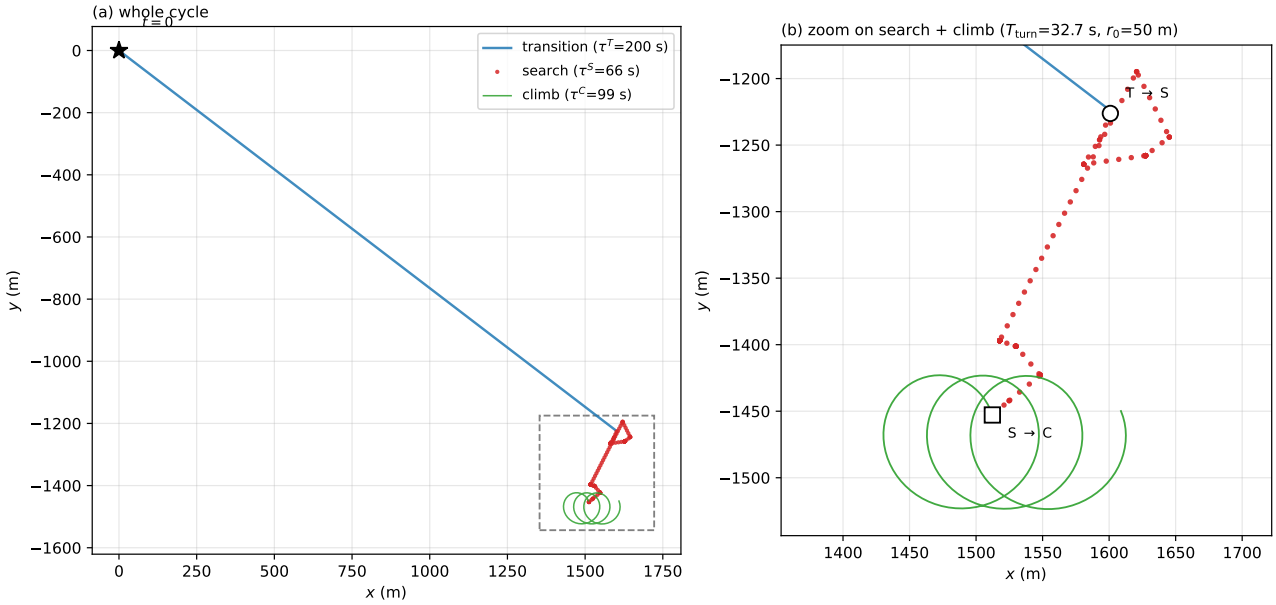

Figure 8: Zoom on one complete soaring cycle of a simulated paraglider, consisting of a single transition phase (blue), search phase (red) and climb phase (green), with phase-boundary markers. The three phases display their characteristic morphologies on their respective native scales: near-rectilinear $v_{xy}\tau^T$ transition step ($\approx 1.7\,\mathrm{km}$), compact red cloud of Mittag-Leffler CTRW search jumps, and circular thermalling loops of radius $r_0 = 50\,\mathrm{m}$ and cycle-dispersed period $T_{\mathrm{turn},n} \sim \mathcal{N}(60, 35^2)\,\mathrm{s}$ during the climb. The cycle closes with a net vertical displacement (gain of altitude) that the 2-D projection cannot show.

thermal soaring (circling radius, typical turn period, thermal geometry [12]) rather than aircraft-specific performance parameters.

Two testable predictions follow, directly addressing the first future direction identified in [5]. First, the horizontal velocity autocorrelation decomposes as $C_{\mathbf{v}}(t) = v_{xy}^2\,\Pi_{\mathrm{TT}}(t)\,\widetilde{C}_\theta(t)$ (Appendix F), where $\Pi_{\mathrm{TT}}(t)$ is the probability that both t' and $t'+t$ fall within transition phases (intermittency factor) and $\widetilde{C}_\theta(t)$ is an angular correlator that decays exponentially on the persistence timescale t_p. Second, the angular autocorrelation $C_\theta(t) = \langle\cos[\theta(t+t') - \theta(t')]\rangle$ is a simple exponential under the model's **(A3)**: any departure from exponential decay (e.g. a stretched-exponential or power-law tail) would indicate a more complex heading-angle process — a finding testable on the raw data of [5]. The Green–Kubo relation [13] $\delta^2(\Delta) = 2\int_0^\Delta (\Delta - t)\,C_{\mathbf{v}}(t)\,\mathrm{d}t$ then unambiguously maps such temporal correlations onto the observed transport scaling.

The model as specified is intentionally minimal. Natural extensions, in decreasing order of immediacy: (i) refining the calibration of the intra-search CTRW parameters (σ_0, τ_0^S, v_c^S) against aircraft-specific scouting patterns extracted from [5]'s Fig. 4e,h, rather than being fixed to common values, would sharpen the match to the conditional MSD of Fig. 3; (ii) replacing the Gaussian angular-increment assumption **(A3)** with a heavy-tailed increment distribution (e.g. Cauchy or stable) would modify the tail of the cycle-to-cycle heading autocorrelation and alter the angular contribution to the MSD; (iii) adding a wind-drift term $\mathbf{v}_{\mathrm{wind}}$ would separate air-relative from ground-relative motion, the third future direction of [5]; (iv) making σ_θ an adaptive function of past search outcomes would allow the model to reproduce the experience-dependent H-increase reported in [5] without additional free parameters. All four are compatible with the present code architecture.

In summary, the minimal cycle-based CTRW model introduced here reproduces the universal sub-ballistic transport scaling of human soaring flights within $\Delta H \lesssim 0.05$ across the three aircraft classes, with the single calibrated angular parameter $\sigma_\theta \simeq 0.35\,\mathrm{rad}$ fixed *analytically* by the $H = 0.88$ isocontour of the phase diagram (Fig. 7), and identifies the origin of the universality in the shared angular-persistence timescale of transition phases. The three-regime structure of the per-phase conditional

MSDs (Fig. 3 of [5]) is recovered naturally from two physically motivated intra-phase ingredients: an intermittent ballistic CTRW for the search (A2.S) in which exponentially-distributed ballistic repositioning manoeuvres alternate with Mittag-Leffler-distributed tight-circling scouting epochs, and a 2D harmonic oscillator plus orographic drift — with cycle-dispersed turn period — for the climb (A2.C). In the present version of the code we tune *only* the paraglider intra-phase parameters so that their conditional MSDs match Fig. 3 of [5]. For the hang-glider and sailplane classes we keep the Vilpellet-reported phase-tail exponents $\mu_{\mathrm{T}}, \mu_{\mathrm{S}}, \mu_{\mathrm{C}}$ and mean climb duration fixed, and tune only the Pareto scales $\tau_0^{\mathrm{T}}, \tau_0^{\mathrm{S}}$ (which are model-introduced parameters and not published in [5]) so that $\langle \tau^{\mathrm{T}} \rangle$ and $\langle \tau^{\mathrm{S}} \rangle$ — and hence the mean cycle duration $\langle T \rangle$ — match the paraglider values. This single tuning knob (per aircraft, per phase) collapses $\delta^2(\Delta)/v_{xy}^2$ onto a single master curve (Fig. 3), reproducing the universal transport reported in the inset of Fig. 1 of [5], and places the three aggregate Hurst exponents in a tight cluster $H = 0.934, 0.930, 0.937$ about a common value $H \simeq 0.93$.

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Appendix A: Survival vs density convention

A Pareto distribution with lower cutoff $\tau_{\min} > 0$ and tail parameter $\mu > 0$ can be characterised through its density,

$$f(\tau) = \mu\, \tau_{\min}^{\mu}\, \tau^{-(\mu+1)}, \qquad \tau \geq \tau_{\min}, \tag{24}$$

or equivalently through its survival function,

$$S(\tau) \equiv P(\tau' > \tau) = (\tau_{\min}/\tau)^{\mu}, \qquad \tau \geq \tau_{\min}. \tag{25}$$

Under this ("*survival-convention*") choice of μ the moments satisfy $\langle \tau \rangle < \infty \iff \mu > 1$ and $\langle \tau^2 \rangle < \infty \iff \mu > 2$, while the density decays with exponent $(\mu+1)$. The alternative ("*density-convention*") choice, $f(\tau) \propto \tau^{-\mu_{\mathrm{d}}}$, corresponds to $\mu_{\mathrm{d}} = \mu + 1$ and shifts each convergence threshold upwards by one.

The authors of [5] explicitly adopt the survival convention in the caption of their Fig. 4 ("tail exponents μ defined through $P(\bullet > \tau) \sim 1/\tau^{\mu}$") but use the density-convention formula $\delta^2 \sim \Delta^{4-\mu}$ and the density-convention statement "$\mu < 3$, infinite variance" in their text. We consistently adopt the declared *survival* convention. Under this choice all three aircraft have $\mu_T > 2$ (Table 1), so $\langle \ell^2 \rangle < \infty$ and the Monte Carlo results of the Monte Carlo section rest on Eq. (23) without Lévy-walk corrections.

Appendix B: Angular correlator

From Eq. (10) the heading difference $\theta_{n+k} - \theta_n = \sum_{j=n+1}^{n+k} \eta_j$ is a sum of k independent $\mathcal{N}(0, \sigma_\theta^2)$ increments, hence itself $\mathcal{N}(0, k\sigma_\theta^2)$ for $k \geq 0$ (for $k < 0$, replace k by $|k|$). The characteristic function of a Gaussian $X \sim \mathcal{N}(0, s^2)$ is $\langle \mathrm{e}^{\mathrm{i}X} \rangle = \mathrm{e}^{-s^2/2}$, so $\langle \cos X \rangle = \mathrm{Re}\, \langle \mathrm{e}^{\mathrm{i}X} \rangle = \mathrm{e}^{-s^2/2}$. With $s^2 = k\sigma_\theta^2$,

$$\langle \cos(\theta_{n+k} - \theta_n) \rangle = \mathrm{e}^{-|k|\sigma_\theta^2/2} \equiv \rho^{|k|}, \quad \rho \equiv \mathrm{e}^{-\sigma_\theta^2/2}. \tag{26}$$

The correlator decays to $1/\mathrm{e}$ at $|k| = N_p = 2/\sigma_\theta^2$, defining the persistence length in cycle units. In physical time $t_p = N_p \langle T \rangle$ provided $\langle T \rangle < \infty$.

Appendix C: Derivation of the MSD formula

Iterating Eq. (1) with $\mathbf{x}_0 = \mathbf{0}$,

$$\mathbf{x}_N = \sum_{k=1}^{N} \ell_k\, \hat{\mathbf{e}}(\theta_k), \tag{27}$$

so that

$$|\mathbf{x}_N|^2 = \sum_{i=1}^{N}\sum_{j=1}^{N} \ell_i\,\ell_j\,\hat{\mathbf{e}}(\theta_i)\cdot\hat{\mathbf{e}}(\theta_j). \tag{28}$$

Taking the expectation and separating the $i = j$ (diagonal) from the $i \neq j$ (off-diagonal) contributions,

$$\begin{aligned}\langle|\mathbf{x}_N|^2\rangle = &\sum_{i=1}^{N}\langle\ell_i^2\rangle\,\langle\hat{\mathbf{e}}(\theta_i)\cdot\hat{\mathbf{e}}(\theta_i)\rangle \\ &+\sum_{i\neq j}\langle\ell_i\ell_j\rangle\,\langle\hat{\mathbf{e}}(\theta_i)\cdot\hat{\mathbf{e}}(\theta_j)\rangle,\end{aligned} \tag{29}$$

having used the independence of step lengths and headings (assumption **(A1)**) to factor the expectations. Because $\hat{\mathbf{e}}(\theta)$ is a unit vector, $\hat{\mathbf{e}}(\theta_i)\cdot\hat{\mathbf{e}}(\theta_i) = 1$, and for $i \neq j$, $\hat{\mathbf{e}}(\theta_i)\cdot\hat{\mathbf{e}}(\theta_j) = \cos(\theta_j - \theta_i)$. By the i.i.d. assumption on ℓ, $\langle\ell_i^2\rangle = \langle\ell^2\rangle$ and $\langle\ell_i\ell_j\rangle = \langle\ell\rangle^2$ for $i \neq j$. Using Eq. (26) and the symmetry of $\rho^{|j-i|}$ under $i \leftrightarrow j$, $\sum_{i\neq j} = 2\sum_{1\leq i<j\leq N}$, and Eq. (29) becomes

$$\langle|\mathbf{x}_N|^2\rangle = N\,\langle\ell^2\rangle + 2\langle\ell\rangle^2 \sum_{1\leq i<j\leq N}\rho^{j-i}, \tag{30}$$

which is Eq. (14).

Appendix D: Evaluation of the geometric sum

At fixed gap $k = j - i$ in the range $1 \leq i < j \leq N$, the pair (i, j) can be chosen in $N - k$ ways (with $i \in \{1, \ldots, N-k\}$ and $j = i + k$), so

$$\sum_{1\leq i<j\leq N}\rho^{j-i} = \sum_{k=1}^{N-1}(N-k)\rho^k. \tag{31}$$

Split the RHS as $N\sum_{k=1}^{N-1}\rho^k - \sum_{k=1}^{N-1}k\rho^k$. Using the standard geometric identities

$$\sum_{k=1}^{N-1}\rho^k = \frac{\rho(1-\rho^{N-1})}{1-\rho}, \tag{32}$$

and

$$\sum_{k=1}^{N-1}k\rho^k = \frac{\rho - N\rho^N + (N-1)\rho^{N+1}}{(1-\rho)^2} \tag{33}$$

(the latter obtained by differentiating $\sum_{k=0}^{N-1}\rho^k = (1-\rho^N)/(1-\rho)$ with respect to ρ and multiplying by ρ), one gets

$$\begin{aligned}\sum_{k=1}^{N-1}(N-k)\rho^k = &\frac{N\rho(1-\rho^{N-1})}{1-\rho} \\ &-\frac{\rho - N\rho^N + (N-1)\rho^{N+1}}{(1-\rho)^2}.\end{aligned} \tag{34}$$

Putting on a common denominator $(1-\rho)^2$ and simplifying the numerator:

$$\begin{aligned}\text{num} &= N\rho(1-\rho)(1-\rho^{N-1}) \\ &\quad -\rho[1 - N\rho^{N-1} + (N-1)\rho^N] \\ &= \rho[\,N(1-\rho) - (1-\rho^N)\,],\end{aligned} \tag{35}$$

so that

$$\sum_{k=1}^{N-1}(N-k)\rho^k = \frac{N\rho}{1-\rho} - \frac{\rho(1-\rho^N)}{(1-\rho)^2}, \tag{36}$$

which is Eq. (15).

Appendix E: Heavy-tail Lévy-walk scaling

For $1 < \mu_T < 2$ in the survival convention one has $\langle \ell \rangle < \infty$ but $\langle \ell^2 \rangle = \infty$. The step distribution lies in the domain of attraction of a symmetric μ_T-stable law, and by the generalised central limit theorem [2] $N^{-1/\mu_T}\sum_{k=1}^{N}(\ell_k - \langle \ell \rangle)$ converges weakly to a symmetric μ_T-stable distribution L_{μ_T} of scale parameter γ_ℓ. In the coupled (finite-velocity) Lévy-walk variant — in which each step of length ℓ also costs a time proportional to ℓ, so that physical time and spatial coordinate are coupled through v_{xy} — this implies

$$\langle |\mathbf{x}_N|^2 \rangle_{\mathrm{LW}} \sim N^{3-\mu_T}, \qquad 1 < \mu_T < 2, \tag{37}$$

which replaces the leading term $N\langle \ell^2 \rangle$ of Eq. (23) when the step-length second moment diverges. The angular contribution — the second term of Eq. (23), proportional to $\langle \ell \rangle^2$ — retains its form as long as $\mu_T > 1$, because it depends only on the step length through its first moment. None of the aircraft analysed in [5] satisfies $1 < \mu_T < 2$ (Table 1), so Eq. (37) is not applicable to the data and Eq. (23) governs the MSD throughout.

Appendix F: Velocity autocorrelation function

Let $\chi_n(t) = 1$ if time t falls within the transition phase of cycle n, and 0 otherwise. Under (A2) the horizontal velocity is

$$\mathbf{v}(t) = v_{xy} \sum_n \chi_n(t)\, \hat{\mathbf{e}}(\theta_n). \tag{38}$$

Since different cycles are disjoint in time, $\chi_n(t')\chi_m(t'+t) = 0$ unless the two indicator functions are active for the appropriate cycles; consequently

$$\begin{aligned} C_{\mathbf{v}}(t) &\equiv \langle \mathbf{v}(t') \cdot \mathbf{v}(t'+t) \rangle \\ &= v_{xy}^2 \sum_{n,m} \langle \chi_n(t')\, \chi_m(t'+t) \rangle\, \rho^{|n-m|}, \end{aligned} \tag{39}$$

where we used the independence of the heading process from the phase indicators (assumption **(A1)**) and Eq. (26). Let $p_{n,m}(t) \equiv \langle \chi_n(t')\chi_m(t'+t) \rangle$ be the joint probability that t' and $t'+t$ belong to the transition phases of cycles n and m respectively, and define

$$\Pi_{\mathrm{TT}}(t) \equiv \sum_{n,m} p_{n,m}(t), \quad \widetilde{C}_\theta(t) \equiv \frac{\sum_{n,m} p_{n,m}(t)\, \rho^{|n-m|}}{\Pi_{\mathrm{TT}}(t)}. \tag{40}$$

Then

$$C_{\mathbf{v}}(t) = v_{xy}^2\, \Pi_{\mathrm{TT}}(t)\, \widetilde{C}_\theta(t). \tag{41}$$

This factorisation is the physical content of the claim in the main text: the velocity autocorrelation is the product of an *intermittency factor* $\Pi_{\mathrm{TT}}(t)$ (the probability that both times fall in transition phases), which decays on the typical cycle duration $\langle T \rangle$, and an *angular correlator* $\widetilde{C}_\theta(t)$, which decays on the persistence timescale $t_p = N_p\langle T \rangle$. The explicit evaluation of Π_{TT} from the phase-duration renewal statistics is left for future work; for the present purposes it is sufficient to note that the decomposition Eq. (41) predicts an exponential decay of $\widetilde{C}_\theta$ — a directly testable prediction on the empirical trajectories of [5]. Combined with the Green–Kubo relation $\delta^2(\Delta) = 2\int_0^\Delta (\Delta - t)\, C_{\mathbf{v}}(t)\, \mathrm{d}t$ [13], this

provides a phase-resolved discriminator between the present model and alternative heading dynamics (e.g. heavy-tailed angular increments or long-memory persistence).

Appendix G: Elementary derivation of the search-phase MSD

This appendix gives the elementary renewal-counting derivation of the conditional search-phase MSD $\delta_S^2(\tau)$ used in Eq. (19); the rigorous continuum formulation in terms of an inverse stable subordinator is the subject of App. 5. The climb-phase MSD derivation is given in the main text (Sec. 3).

Search (A2.S): intermittent ballistic CTRW with ML rests. The pilot performs a sequence of ballistic legs of duration $\sigma_k \sim \mathrm{Exp}(\sigma_0)$ at speed v_c^S in directions ψ_k, interleaved with waiting epochs of duration $\xi_k \sim \mathrm{ML}(\alpha_S, \tau_0^S)$, $\alpha_S \in (0,1)$, during which the horizontal position is stationary at the GPS sampling scale. Leg directions evolve by a persistent Gaussian random walk on the circle, so the leg-displacement vectors $\Delta\mathbf{r}_k = v_c^S \sigma_k \hat{\mathbf{e}}(\psi_k)$ are isotropic on average ($\langle \hat{\mathbf{e}}(\psi_k) \cdot \hat{\mathbf{e}}(\psi_{k'}) \rangle \to 0$ for $|k-k'|$ large).

At physical time t, the cumulative displacement is $\mathbf{X}^S(t) = \sum_{k:t_k \le t} \Delta\mathbf{r}_k$ where t_k are leg *end* times (legs prior to t_k are complete; the leg that is currently in progress at t contributes a fractional ballistic advance). By the isotropy of the leg direction process in the ensemble,

$$\langle |\mathbf{X}^S(t)|^2 \rangle \simeq (v_c^S)^2 \langle \sigma_k^2 \rangle \, \langle N(t) \rangle = 2(v_c^S \sigma_0)^2 \langle N(t) \rangle, \tag{42}$$

where $N(t)$ is the number of completed legs in $[0,t]$ and we used $\langle \sigma_k^2 \rangle = 2\sigma_0^2$ for an exponential leg-duration law. The mean number of renewals of a CTRW with Mittag-Leffler waiting times is, for $\tau_0^S \ll t$ [1, 8],

$$\langle N(t) \rangle \simeq \frac{(t/\tau_0^S)^{\alpha_S}}{\Gamma(1+\alpha_S)}, \tag{43}$$

so the search-phase MSD scales asymptotically as

$$\delta_S^2(t) \simeq \frac{2\,(v_c^S \sigma_0)^2}{\Gamma(1+\alpha_S)} \left(\frac{t}{\tau_0^S} \right)^{\alpha_S}, \qquad t \gg \tau_0^S, \tag{44}$$

reproducing the third regime of Eq. (5). At short lags $t \ll \sigma_0$ the MSD is ballistic, $\delta_S^2(t) \simeq (v_c^S)^2 t^2$, from the motion inside a single leg. The intermediate regime $\sigma_0 \ll t \ll \tau_0^S$ is a finite-size crossover whose effective exponent depends on the ratio σ_0/τ_0^S (for the paragliders configuration $\sigma_0 = \tau_0^S = 3\,\mathrm{s}$ so this regime is narrow and the crossover is essentially immediate). The characteristic displacement scale of a single scouting step is $v_c^S \sigma_0 \sim 57\,\mathrm{m}$ for paragliders ($v_c^S = 19\,\mathrm{m/s}$, $\sigma_0 = 3\,\mathrm{s}$), much smaller than the transition step $v_{xy}\langle \tau^T \rangle \sim 2 \times 10^3\,\mathrm{m}$; this separation of scales is why the search contribution to the total MSD is a sub-leading correction to the transition contribution.

The derivation of the climb-phase MSD (A2.C), including the Gaussian damping of the coherent oscillations by the per-cycle T_{turn} dispersion and the resulting three regimes of the green curves of Fig. 6, is given in the main text (Sec. 3, paragraph *"Climb MSD and the three regimes of the green curve"*), since it is the key quantitative prediction for the climb phase and is used directly in the interpretation of the data.

Appendix H: Subordination representation of the search-phase CTRW

This appendix derives Eq. (4) and, in doing so, makes the subordinator that enters the search-phase dynamics explicit. The search-phase motion (A2.S) is a two-state renewal process with alternating *active* intervals $\sigma_k \sim \mathrm{Exp}(\sigma_0)$ (during which the walker translates ballistically at speed v_c^S) and *waiting* intervals $\xi_k \sim \mathrm{ML}(\alpha_S, \tau_0^S)$ with $\alpha_S \in (0,1)$. We derive (i) the mean number of completed legs $\langle N(t) \rangle$, (ii) its continuum limit in terms of an inverse stable subordinator, and (iii) the resulting scaling of the search-phase MSD.

Setup and Laplace transforms. Let $T_k = \sigma_k + \xi_k$ be the kth inter-event interval. The T_k are i.i.d. and the counting process

$$N(t) \;=\; \max\{n \geq 0 : \textstyle\sum_{k\leq n} T_k \leq t\}$$

is the number of completed legs in $[0, t]$. Because the Mittag–Leffler ξ_k has infinite mean, so does T_k; the classical (finite-mean) elementary renewal theorem does not apply and $\langle N(t)\rangle$ is sub-linear. Using the standard Laplace-transform identity for renewal densities $m(t) \equiv d\langle N(t)\rangle/dt$,

$$\hat{m}(s) \;=\; \frac{\hat{f}_T(s)}{1-\hat{f}_T(s)}, \qquad \hat{f}_T(s) \;=\; \hat{f}_\sigma(s)\,\hat{f}_\xi(s). \tag{45}$$

The exponential leg has $\hat{f}_\sigma(s) = 1/(1+\sigma_0 s)$, with the small-$s$ expansion $\hat{f}_\sigma(s) = 1 - \sigma_0 s + O(s^2)$. The Mittag–Leffler wait has the characteristic small-s behaviour

$$\hat{f}_\xi(s) \;=\; \frac{1}{1+(\tau_0^S s)^{\alpha_S}} \;=\; 1 - (\tau_0^S s)^{\alpha_S} + O(s^{2\alpha_S}), \quad s \to 0^+, \tag{46}$$

which encodes the heavy tail through the non-analytic s^{α_S} term. Inserting both expansions into Eq. (45),

$$1 - \hat{f}_T(s) \;=\; (\tau_0^S s)^{\alpha_S} + \sigma_0 s + O(s^{2\alpha_S}),$$

and the s^{α_S} term dominates for $s \to 0$ whenever $\alpha_S < 1$. Therefore

$$\hat{m}(s) \;\sim\; \frac{1}{(\tau_0^S s)^{\alpha_S}}, \quad s \to 0^+. \tag{47}$$

Inverse Laplace transform of $s^{-\alpha_S}$ gives $t^{\alpha_S - 1}/\Gamma(\alpha_S)$, so

$$\langle N(t)\rangle \;=\; \int_0^t m(t')\,dt' \;\sim\; \frac{1}{\Gamma(1+\alpha_S)}\left(\frac{t}{\tau_0^S}\right)^{\alpha_S}, \quad t \gg \tau_0^S, \tag{48}$$

which is the sub-linear renewal law of the Mittag–Leffler CTRW.

Continuum limit: the inverse stable subordinator. The cumulative waiting-time process $\Xi_n \equiv \sum_{k=1}^n \xi_k$, rescaled by n^{1/α_S}, converges (generalised central limit theorem for sums of α_S-stable random variables [2]) to a strictly-increasing one-sided α_S-stable Lévy motion $S_{\alpha_S}(\tau)$ — the *α_S-stable subordinator*. By the Meerschaert–Scheffler limit theorem [9] its inverse

$$E_{\alpha_S}(t) \;\equiv\; \inf\{\tau \geq 0 : S_{\alpha_S}(\tau) > t\} \tag{49}$$

(the *inverse α_S-stable subordinator*, also known as the Mittag–Leffler process) is the continuum limit of the rescaled counting process $N(t)/\mathrm{const}(\tau_0^S) \Rightarrow E_{\alpha_S}(t)$. Its mean reproduces Eq. (48), $\langle E_{\alpha_S}(t)\rangle = t^{\alpha_S}/\Gamma(1+\alpha_S)$, and its density satisfies a fractional Fokker–Planck equation of order α_S [1, 8].

Subordination of the displacement. Let $\tilde{\mathbf{X}}^S(n) \equiv \sum_{k=1}^n v_c^S \sigma_k \hat{\mathbf{e}}_k$ denote the *leg-count-indexed* displacement accumulated after n completed legs. The isotropy of the leg-direction process gives, for large n, $\langle|\tilde{\mathbf{X}}^S(n)|^2\rangle = n\,\langle\sigma_k^2\rangle(v_c^S)^2 = 2n(v_c^S\sigma_0)^2$. The physical-time displacement of the walker at time t is obtained by time-changing the leg counter $n \mapsto N(t)$,

$$\mathbf{X}^S(t) \;=\; \tilde{\mathbf{X}}^S\big(N(t)\big) \;+\; \big[\text{fractional advance along the in-progress leg}\big]. \tag{50}$$

The fractional term is bounded uniformly by $v_c^S\sigma_0$ and is negligible at $t \gg \sigma_0$. In the continuum limit $N(t) \to E_{\alpha_S}(t)$ and $\tilde{\mathbf{X}}^S(\cdot)$ tends to a Brownian motion (since σ_k has finite second moment and the heading process is mixing), so Eq. (50) reduces to Eq. (4), $\mathbf{X}^S(t) \overset{d}{=} \tilde{\mathbf{X}}^S(E_{\alpha_S}(t))$, which is the standard time-change (subordination) representation of a Mittag–Leffler CTRW [8].

MSD implication. Taking expectations in Eq. (50) and using isotropy,

$$\delta_S^2(t) \;\simeq\; 2(v_c^S\sigma_0)^2\,\langle N(t)\rangle \;\sim\; \frac{2(v_c^S\sigma_0)^2}{\Gamma(1+\alpha_S)}\left(\frac{t}{\tau_0^S}\right)^{\alpha_S}, \quad t \gg \tau_0^S. \tag{51}$$

This recovers the third regime of Eq. (5) and identifies the origin of the sub-diffusive exponent α_S with the order of the inverse stable subordinator E_{α_S}. Physically: the walker accumulates ballistic displacement in an *operational* clock that runs proportional to t^{α_S}, not t, because the heavy-tailed rests ξ_k progressively slow the event rate as t grows.

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